

TOPIC NAME : _____

DAY : _____

TIME : _____ DATE : / /

MOHSIN IBNA HOSSAIN

AIUB - MATH - 1 - NOTE

MATH - 1

TOPIC NAME : _____

DAY : _____

TIME : _____ DATE : / /

Table of Contents

SL.NO	Topic	Pages
1	Function	03-24
2	Limit	25-32
3	Continuity	33-38
4	Differentiation	39-48
5	Implicitly Differentiation	49-58
6	Application of Differentiation	59-67
7	Taylor & Maclaurin Series	68-75
8	Analysis of Function	76-102
9	Optimization Problem	103-106

Functions!

Function:

a) Area, $A = \pi r^2$
 $A \propto r^2$

$r = 1 ; A = 1$

$r = 2 ; A = 4$

$f(n) = n^2$

$f(1) = 1^2$

$f(2) = 2^2$

Number line:



$[1, 4] \rightarrow 1 \leq n \leq 4$

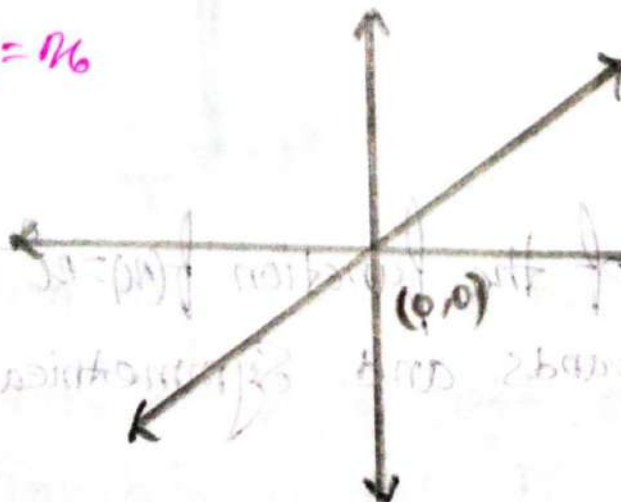
$(1, 4] \rightarrow 1 < n \leq 4$

$[1, 4) \rightarrow 1 \leq n < 4$

Sketch the graph:

Power function:

$y = f(n) = n$

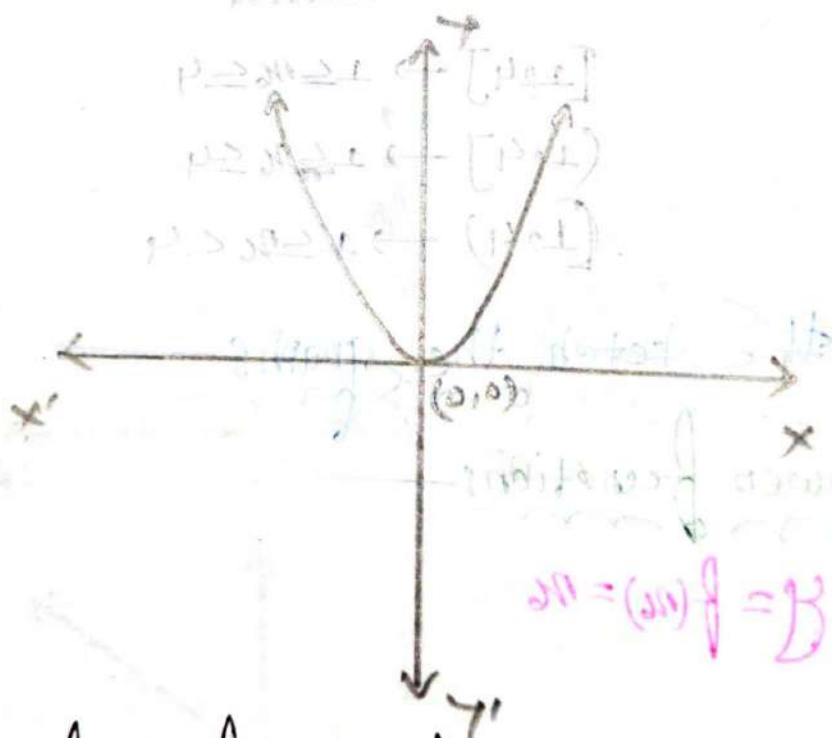


$\Rightarrow$ The graph of the function $f(x) = x$ is a Straight line, that passes through the origin $(0,0)$ and has a Slope of 1.

if $x \geq 1$ then $f(x) = 1$
 $x \leq -1$ then $f(x) = -1$

the line represents a one to one relationship between x and $f(x)$, meaning that for every value of x there is a Unique value of $f(x)$, and vice versa

② $y = x^2$



$\Rightarrow$ The graph of the function $f(x) = x^2$ is a Parabola. that opens upwards and Symmetrical about the x-axis.

The Vertex of the Parabola is at the origin $(0, 0)$

Q:- What is the name of following functions and sketch it?

* $y = (x+2)^2$

Let, $y = f(x)$

and $x+2 = x_0$

So that, function is $f(x_0) = x_0^2$

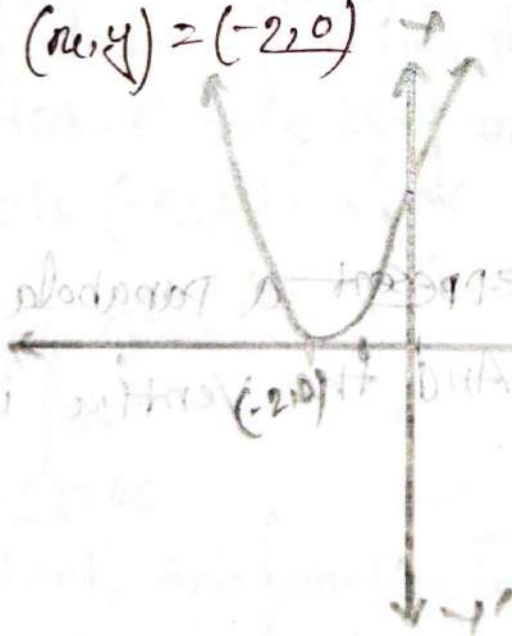
if $x_0 = 0$

then $x+2 = 0$

$x = -2$

and $y = (-2+2)^2$
 $= 0$

$\therefore (x, y) = (-2, 0)$



This given function represent a Parabola which is shifted two units of 2 on left side and the Vertex of the Parabola is $(-2, 0)$.

$$* y = (x-2)^2$$

$$\Rightarrow \text{Let } y = f(x)$$

$$(x-2) = x$$

then, the function is $f(x) = x^2$

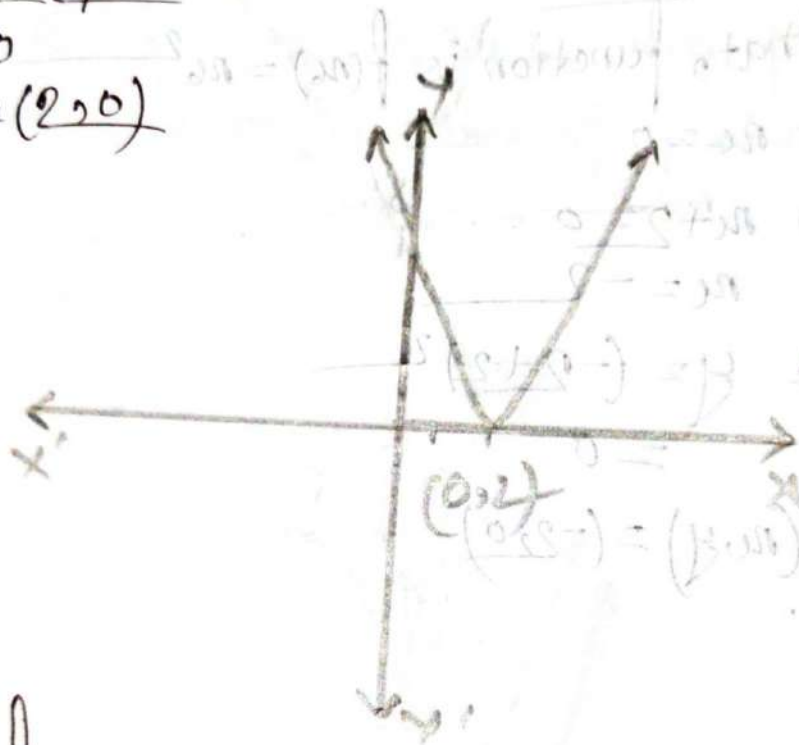
$$\text{if } x = 0$$

$$\text{so } x-2 = 0$$

$$\text{or } x = 2$$

$$\text{and } y = (2-2)^2$$

$$\text{So } (x, y) = (2, 0)$$



$\rightarrow$ this given function represent a parabola. which is shifted 2 units on right side. And the vertex is $(2, 0)$

$$* y = (x+2)^2 + 1$$

$$\Rightarrow \text{Let, } y = f(x)$$

$$(x+2) = x$$

then the function is $f(x) = x^2 + 1$

if $x=0$

$$\text{So } x+2=0$$

$$\text{on } x=-2$$

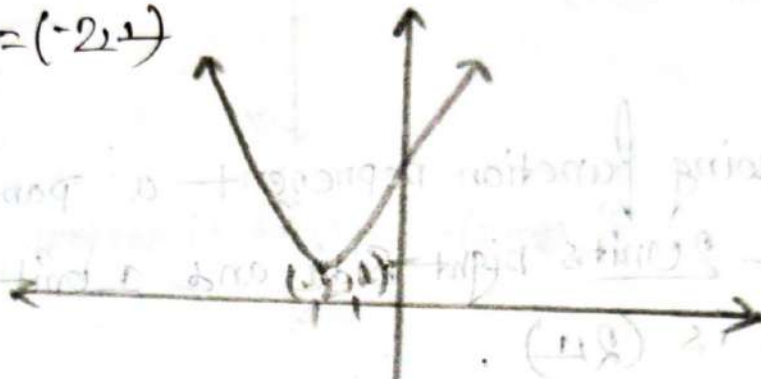
$$\text{and } y = (x+2)^2 + 1$$

$$= (-2+2)^2 + 1$$

$$= 0 + 1$$

$$= 1$$

$$\text{So, } (x, y) = (-2, 1)$$



→ The following function represent a parabola, which is shifted 2 units left and 1 units top side. And its vertex is $(-2, 1)$.

$$* y = (x-2)^2 + 1$$

$$\Rightarrow \text{Let, } y = f(x)$$

$$(x-2)=x$$

So that, the function is $f(x) = x^2 + 1$

$$\text{if } x=0$$

$$\text{So } x-2=0$$

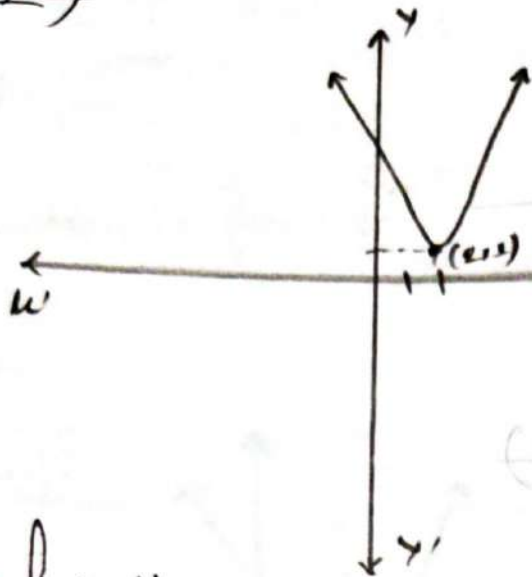
$$\text{on } x=2$$

$$\text{and } y = (x-2)^2 + 1$$

$$= (2-2)^2 + 1$$

$$= 0 + 1 = 1$$

So, $(n, y) = (2, 1)$



→ the following function represent a parabola. which is shifted 2 units right side and 1 unit top side. and the vertex is $(2, 1)$.

Q. $y = -x^2$

Let, $y = f(x)$

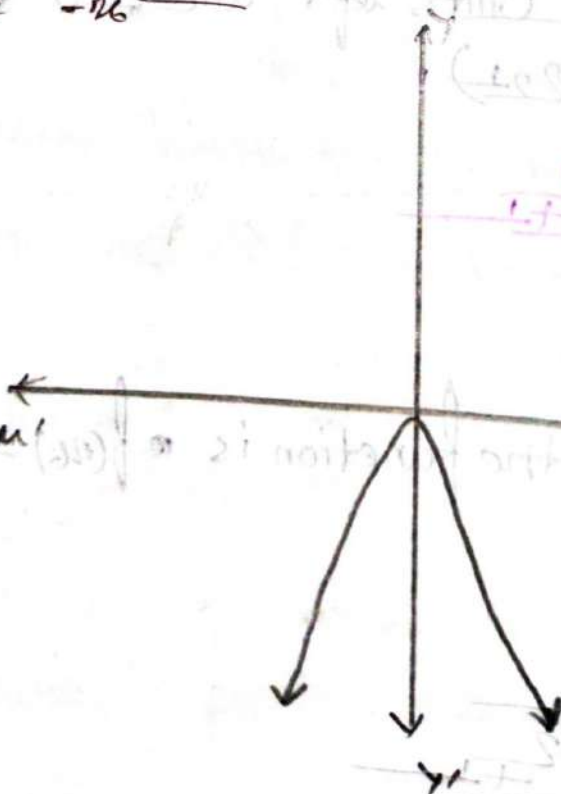
$x = 0$

function is $f(x) = -x^2$

if $x = 0$

then $y = 0$

So, $(n, y) = (0, 0)$



⇒ The graph of the function $f(x) = -x^2$ is a parabola that opens downwards and is symmetrical about the x-axis. The vertex of parabola is at the origin (0,0).

Q8 What is the name of a following function and sketch it.

* $y = -(x+2)^2$

Let, $y = f(x)$

$x+2 = x$

So that the function is $f(x) = 1 - (x)^2$

if $x=0$

or, $x+2 = 0$

or, $x = -2$

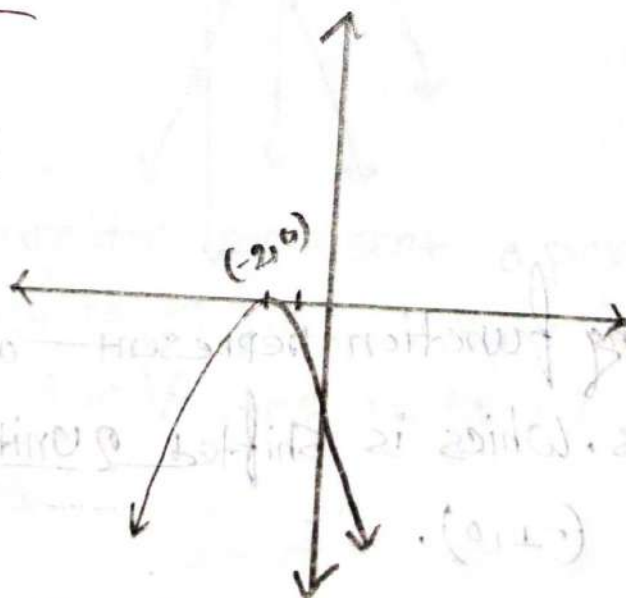
and $y = -(x+2)^2$

$= -(-2+2)^2$

$= -(0)^2$

$= 0$

So $(x, y) = (-2, 0)$



⇒ the following function is represent a parabola that opens downwards. which is shifted 2 units on left side and the vertex is $(-2, 0)$.

Q8: * $y = -(x+1)^2$

Let, $y = f(x)$

$x+1 = x$

So, the function is $f(x) = -x^2$

if $x = 0$

then; $x+1 = 0$

or, $x = -1$

any $y = f(x)$
 $= -(-1)^2$
 $= 0$

So, $(x, y) = (-1, 0)$



→ the following function represent a Parabola. that opens downwards. which is shifted 2 units on left, and the vertex is $(-1, 0)$.

$$* y = -(x-1)^2 + 1$$

Let, $y = f(x)$

$$x-1 = x$$

So the function is $f(x) = (x-1)^2 + 1$

if, $x = 0$

then $x-1 = 0$

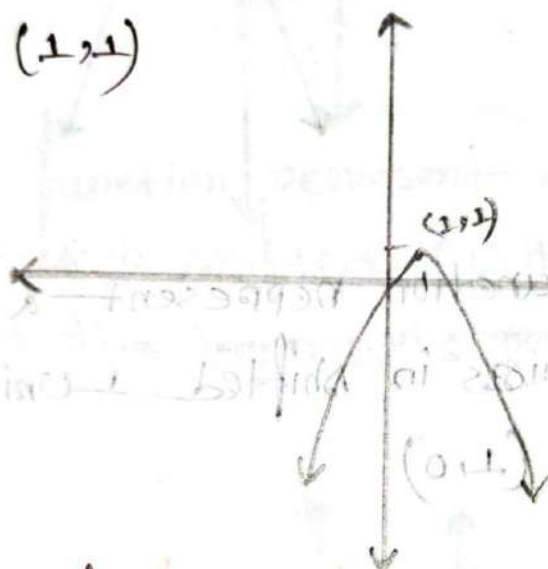
$$x = 1$$

and $y = -(x-1)^2 + 1$

$$= -(1-1)^2 + 1$$

$$= 1$$

So, $(x, y) = (1, 1)$



→ The following function represent a parabola, that opens downwards, which is shifted 1 units up and 1 units right side, and the Vertex is $(1, 1)$

$$* y = -x^2$$

Let, $y = f(x)$

$$x-1 = x$$

So, function is $f(x) = -x^2$

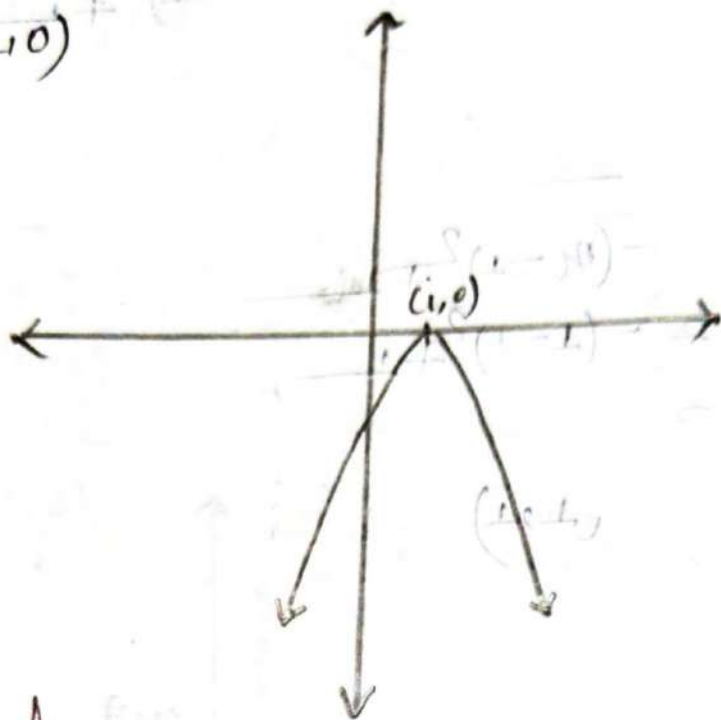
if, $x = 0$

$$x-1 = 0$$

$$\text{on } x = 1$$

$$\begin{aligned} \text{and } y &= -(x-1)^2 \\ &= -(1-1)^2 \\ &= 0 \end{aligned}$$

$$\text{So, } (x, y) = (1, 0)$$



⇒ The following function represent a parabola that opens downwards, which is shifted 1 unit right side, and the vertex is $(1, 0)$

Q Sketch the graph of the function $f(x) = x^2 + 6x + 10$

$$\begin{aligned} f(x) &= x^2 + 6x + 10 \\ &= x^2 + 2 \cdot x \cdot 3 + 3^2 + 1 \\ &= (x+3)^2 + 1 \end{aligned}$$

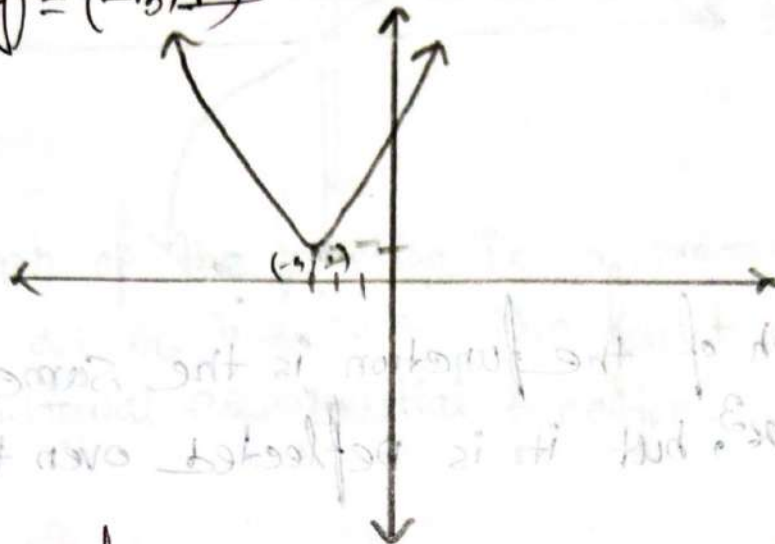
Let, $x+3 = u$, $f(u) = y$

So, the function is $f(u) = u^2 + 1$

if $u+3=0$
 $u = -3$

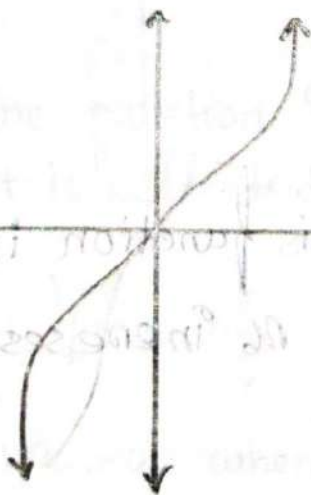
and $y = (x+3)^2 + 1$
 $= (-3+3)^2 + 1$
 $= 1$

So, $(x, y) = (-3, 1)$



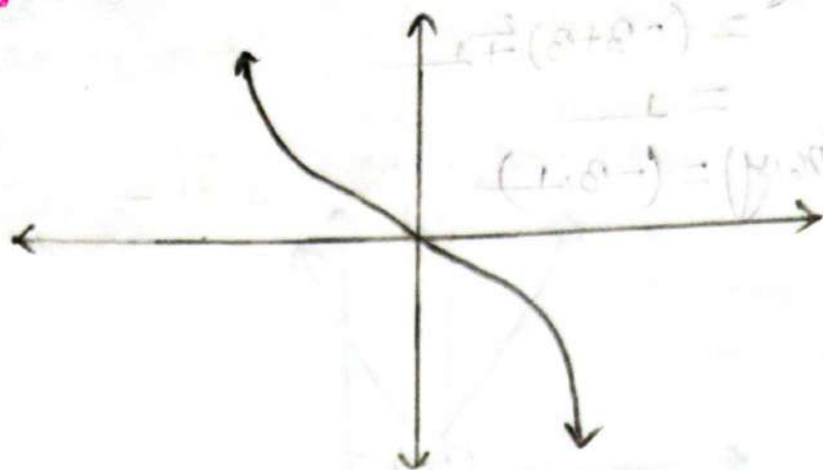
⇒ The following function represent a parabola, that opens Upwards, which is shifted 3 units left and 1 units Top Side, and the Vertex is $(-3, 1)$.

⑤ $y = x^3$



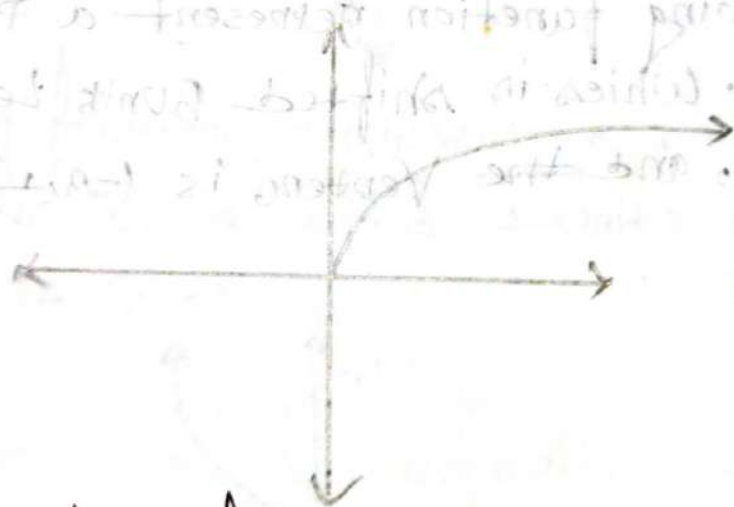
⇒ The following function represent a Curve. That increases or decreases depending on the sign of x .

⑤ $y = -x^3$



⇒ The graph of the function is the same as the graph of $f(x) = x^3$, but it is reflected over the x -axis.

⑥ $y = \sqrt{x}$



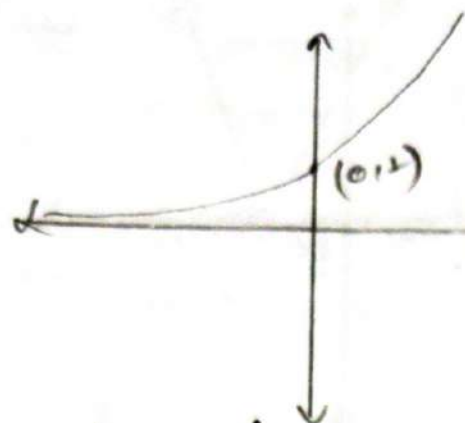
⑦ $y = x^2$

⇒ The graph of this function is a smooth curve that increases as x increases.

⇒ The following function represents a smooth curve that increases as x increases.

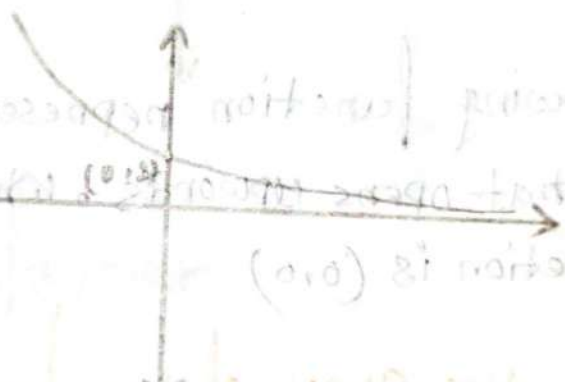
29 13 14

7) $y = e^{x_6}$



$\Rightarrow$ The graph of the function is a smooth curve that increases as x_6 increases. The function is also known as the natural exponential function.

8) $y = -e^{x_6}$



$\Rightarrow$ The graph of the function is the same as the graph of $f(x) = e^x$, but it is reflected over the x -axis.

Absolute Value of function:

$$f(x_6) = \begin{cases} x_6; & x_6 \geq 0 \\ -x_6; & x_6 < 0 \end{cases}$$

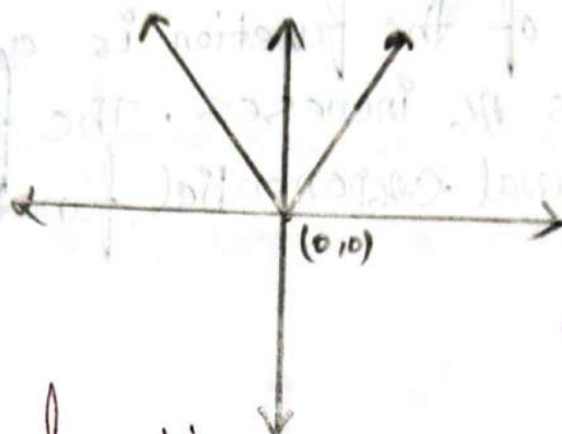
when x_6 is ≥ 0 , then $f(x_6) = x_6$
 when x_6 is < 0 , then $f(x_6) = -x_6$

① $y = |x|$

Let $f(x) = y$
 $x_0 = x_1$

Now, $x_0 = 1$
 $f(x) = 1$

on, $x_0 = -1$
 $f(x) = -1$



⇒ The following function represent a Absolute Value function. that opens upwards, whose are the vertex of the function is (0,0)

Q:- Sketch the graph of the function

① $y = |x+2|$

Let, $y = f(x)$

$x_0 + 2 = x_1$

Now, function is $f(x) = |x+2|$

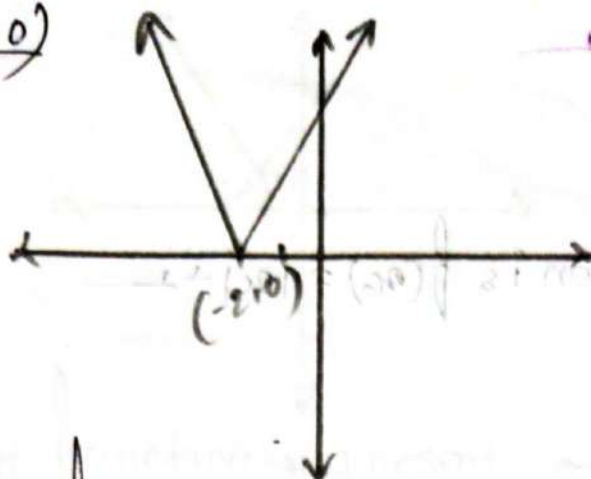
if $x_0 = 0$

so $x_0 + 2 = 2$

on $x_0 = -2$

and $y = 0$

So, $(x, y) = (-2, 0)$



⇒ the following function represent a Absolute Value function. that opens upwards, which is shifted 2 units on left. and the vertex is $(-2, 0)$

Ans

* $y = |x - 2|$

Let $y = f(x)$

$x - 2 = x$

So the function is $f(x) = |x|$

if $x = 0$

so $x - 2 = 0$

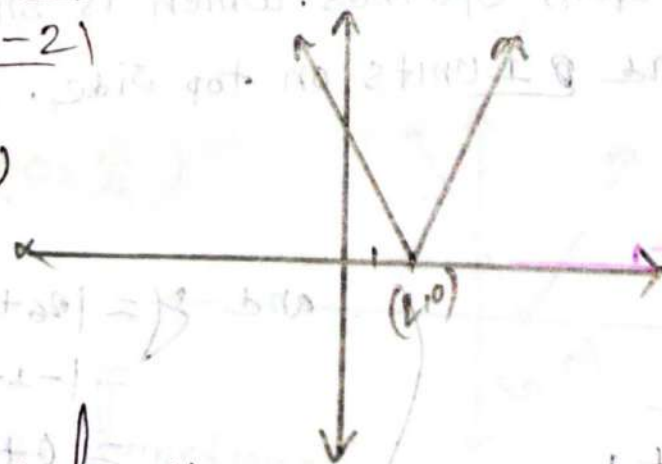
or $x = 2$

and $y = |x - 2|$

$= |2 - 2|$

$= 0$

so, $(x, y) = (2, 0)$



⇒ The following functions represent a absolute value function. that opens upwards, whis is shifted 2 unit on side and the vertex is $(2, 0)$

$$* y = |x-2| + 1$$

Let, $y = f(x)$

$$x-2 = x$$

So the function is $f(x) = |x| + 1$

if $x = 0$

so, $x-2 = 0$

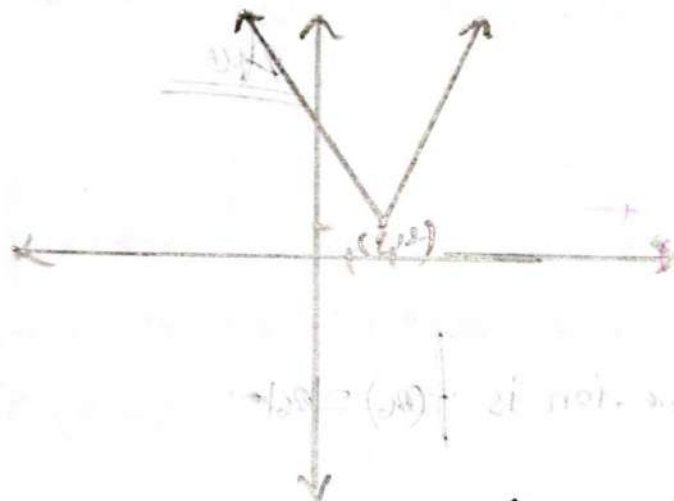
on $x = 2$

and $y = |x-2| + 1$

$$= |2-2| + 1$$

$$= 1$$

$(x, y) = (2, 1)$



⇒ the following function represent a Absolute value function. that opens upwards. which is shifted 2 units on right side and 1 units on top side. and the vertex is $(2, 1)$.

$$* y = |x+1| + 1$$

Let, $f(x) = y$

$$x = x+1$$

so, $f(x) = |x| + 1$

if $x = 0$

$x+1 = 0$

$x = -1$

and $y = |x+1| + 1$

$$= |-1+1| + 1$$

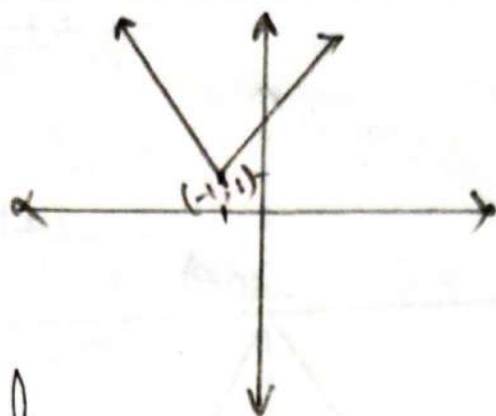
$$= 0 + 1$$

$$= 1$$

18

17

$$\therefore (x, y) = (-1, 1)$$



$$f(x) = |x + 1| + 1$$

⇒ The following function represent a Absolute value function. that open Upwards. which is Shifted 1 unit ~~to~~ left and 1 Unit Top side. and the Vertex is $(-1, 1)$

$$(47) \quad g(t) = |1 - 3t|$$

Let, $g(t) = f(n_6)$

$$1 - 3t = n_6$$

So, $f(n_6) = |n_6|$

if $n_6 = 0$

So $1 - 3t = 0$

On $3t = 1$

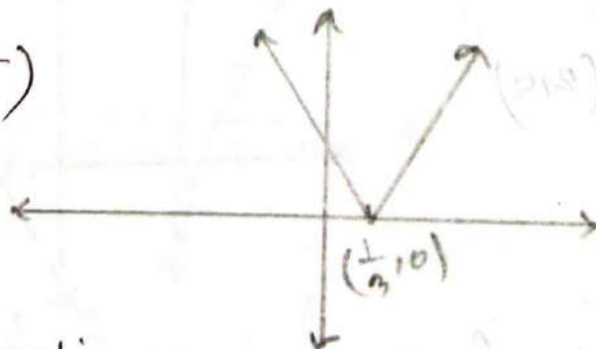
on $t = \frac{1}{3}$

and $g(t) = |1 - 3 \times \frac{1}{3}|$

$$= |1 - 1|$$

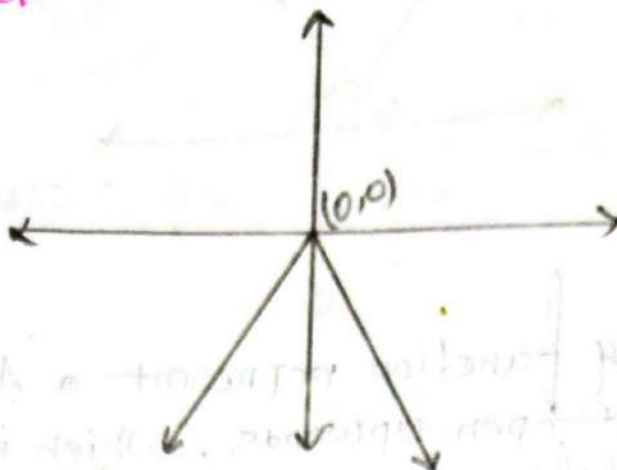
$$= 0$$

$$(g(t), t) = (0, \frac{1}{3})$$



⇒ The following functions represent a absolute value function. that opens Upwards. which is Shifted $\frac{1}{3}$ unit on Right side. the vertex is $(\frac{1}{3}, 0)$

20. $y = -|x|$



⇒ the following function of the graph is a absolute value function, that opens downwards, and the vertex is $(0,0)$.

Qr Sketch the graph of the function:

21. $y = -|x-2|$

Let $y = f(x)$

$x-2 = x$

So $f(x) = -|x|$

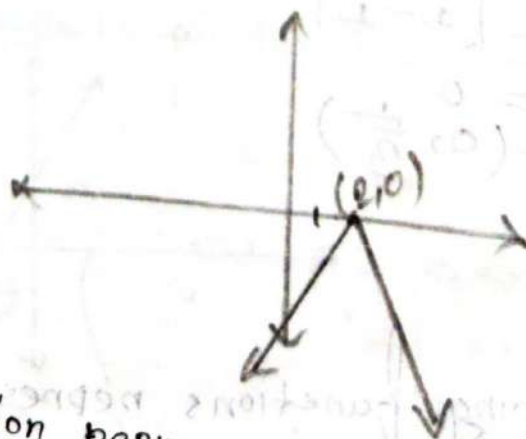
if $x = 0$

So $x-2 = 0$

on $x = 2$

and $y = -|2-2|$
 $= 0$

So $(x,y) = (2,0)$



⇒ The following function represent a absolute value function that opens downwards, which is shifted 2 units on Right Side and the vertex is $(2,0)$

$$* y = -|x+2|+1$$

Let, $y = f(x)$

$$x+2 = x$$

So, $f(x) = -|x|+1$

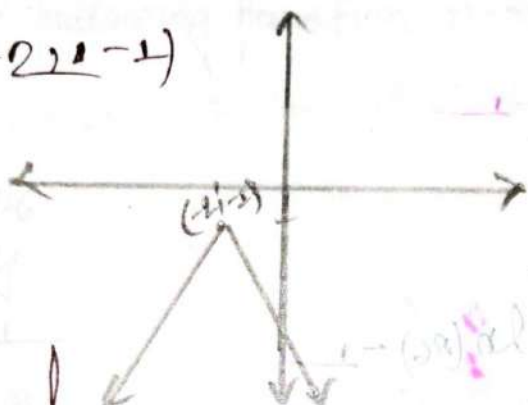
if $x = 0$

or $x+2 = 0$

$$x = -2$$

and $y = -|-2+2|+1$
 $= -1$

$$\therefore (x, y) = (-2, -1)$$

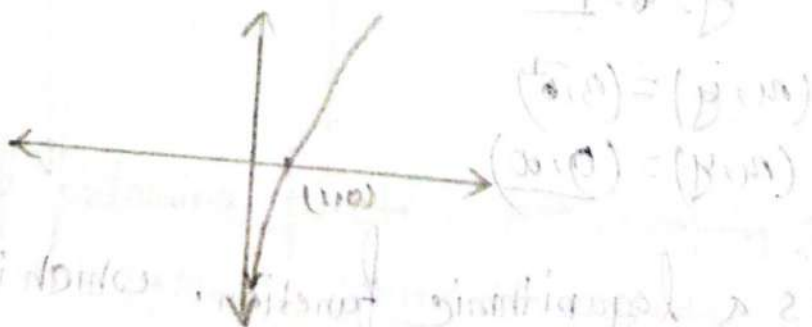


⇒ The following function represent a absolute value function. that opens downwards. which is shifted 2 units on left and 1 units down side. and the vertex is $(-2, -1)$.

ii) $y = \ln(x)$

$$x = 1$$

$$y = 0$$

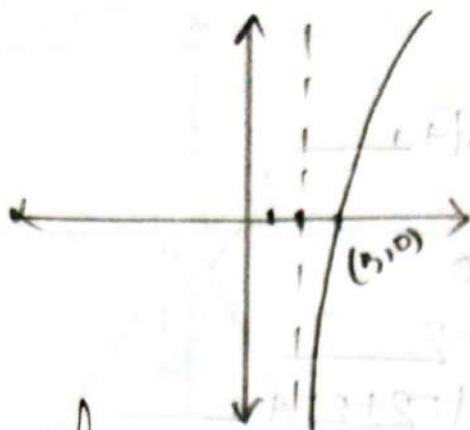


⇒ It's a type of a logarithmic function.

* $y = \ln(x-2)$

$x_0 = 3$

$y = 0$



→ It's a logarithmic function, which is shifted 3 units on Right side.

* $y = \ln(x-2) - 1$

Let $f(x) = y$

$(x-2) = x_0$

So, $f(x) = \ln(x_0) - 1$

if $x_0 - 2 = 0$

$x_0 = 2$

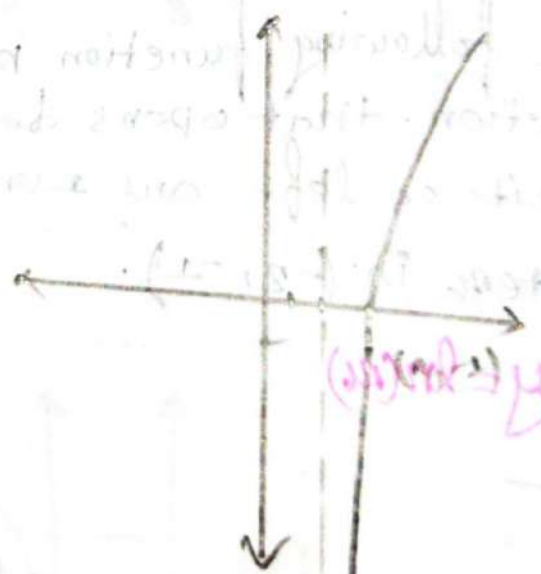
So $y = \ln(x-2) - 1$
 $\rightarrow \infty$

and $x_0 = 3$

$y = -1$

$(x, y) = (3, -1)$

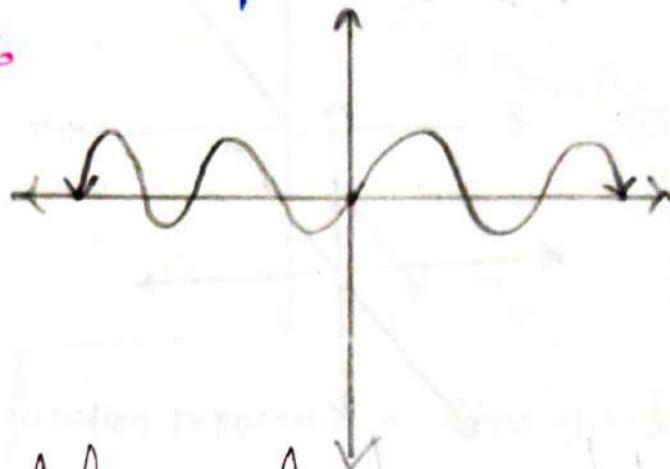
$(x, y) = (2, \infty)$



→ It's a logarithmic function, which is shifted 3 units on Right side and 1 units on bottom side.

Trigonometric functions

(12) $y = \sin \theta$



$\Rightarrow$ The graph of following function represent a Smooth Curve.

(13) $y = 1 - \sin \theta$

Let, $f(\theta) = y$

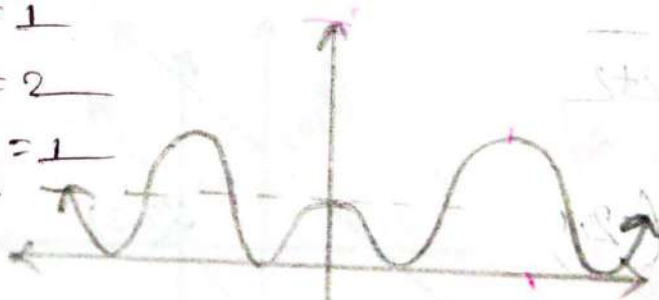
now, $\theta = 0, y = 1$

$\theta = \frac{\pi}{2}, y = 0$

$\theta = \pi, y = 1$

$\theta = \frac{3\pi}{2}, y = 0$

$\theta = 2\pi, y = 1$



$\Rightarrow$ The graph of following function represent a Smooth Curve. which is shifted 1 units up words

Official

Limit

$$\lim_{n_k \rightarrow a} f(n_k) = L$$

when n_k approaches to a then $f(n_k) = L$

Page 8.102:

$$\lim_{n \rightarrow 5} \frac{n^2 - 6n + 5}{n - 5}$$

$$= \lim_{n \rightarrow 5} \frac{n^2 - 5n - n + 5}{n - 5}$$

$$= \lim_{n \rightarrow 5} \frac{n(n-5) - 1(n-5)}{n-5}$$

$$= \lim_{n \rightarrow 5} \frac{(n-1)(n-5)}{(n-5)}$$

$$= \frac{5-1}{1}$$

$$= 4$$

Ans

$$\lim_{h \rightarrow 0} \frac{(-5+h)^2 - 25}{h}$$

$$= \lim_{h \rightarrow 0} \frac{(h-5)^2 - (5)^2}{h}$$

$$= \lim_{h \rightarrow 0} \frac{(h-5-5)(h-5+5)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{h(h-10)}{h}$$

$$= \lim_{h \rightarrow 0} (h-10)$$

$$\left. \begin{aligned} (0+1-10) &= 0-10 \\ (0+1-10) &= -10 \end{aligned} \right\} \text{Ans}$$

$$\star \lim_{h \rightarrow 0} \frac{\sqrt{0+h} - 3}{h}$$

$$\Rightarrow \lim_{h \rightarrow 0} \frac{(\sqrt{0+h} - 3) \cdot (\sqrt{0+h} + 3)}{h(\sqrt{0+h} + 3)}$$

$$= \lim_{h \rightarrow 0} \frac{(\sqrt{0+h})^2 - (3)^2}{h(\sqrt{0+h} + 3)}$$

$$= \lim_{h \rightarrow 0} \frac{0+h-9}{h(\sqrt{0+h} + 3)}$$

$$= \lim_{h \rightarrow 0} \frac{h}{h(\sqrt{0+h} + 3)}$$

$$= \frac{1}{\sqrt{0+0} + 3}$$

$$= \frac{1}{6}$$

Ans:-

$$\textcircled{22} \lim_{u \rightarrow 2} \frac{\sqrt{4u+1} - 3}{u-2}$$

$$= \lim_{u \rightarrow 2} \frac{(\sqrt{4u+1} - 3)(\sqrt{4u+1} + 3)}{(u-2)(\sqrt{4u+1} + 3)}$$

$$= \lim_{u \rightarrow 2} \frac{(\sqrt{4u+1})^2 - (3)^2}{(u-2)(\sqrt{4u+1} + 3)}$$

$$= \lim_{u \rightarrow 2} \frac{4u+1-9}{(u-2)(\sqrt{4u+1} + 3)}$$

30

30/01/23
ms. g. Lec: 03

$$\begin{aligned} &= \lim_{u \rightarrow 2} \frac{4u-8}{(u-2)(\sqrt{4u+1}+3)} \\ &= \lim_{u \rightarrow 2} \frac{4(u-2)}{(u-2)(\sqrt{4u+1}+3)} \\ &= \lim_{u \rightarrow 2} \frac{4}{\sqrt{4u+1}+3} \\ &= \frac{4}{\sqrt{4 \times 2 + 1} + 3} \\ &= \frac{4}{3+3} \\ &= \frac{4}{6} \quad \underline{\underline{\text{Ans: -}}} \end{aligned}$$

Page: 138:

$$\begin{aligned} &\textcircled{15} \lim_{n \rightarrow \infty} \frac{3n-2}{2n+1} \\ &= \lim_{n \rightarrow \infty} \frac{3n-2}{\frac{2n+1}{n}} \\ &= \lim_{n \rightarrow \infty} \frac{3 - \frac{2}{n}}{2 + \frac{1}{n}} \quad \left[\text{As } \lim_{n \rightarrow \infty} \frac{1}{n} = 0 \right] \\ &= \lim_{n \rightarrow \infty} \frac{3-0}{2+0} \\ &= \frac{3}{2} \\ &\quad \underline{\underline{\text{Ans: -}}} \end{aligned}$$

23

$$\lim_{n \rightarrow \infty} (\sqrt{n^2+n} - 3n)$$

$$= \lim_{n \rightarrow \infty} \frac{(\sqrt{n^2+n} - 3n)(\sqrt{n^2+n} + 3n)}{\sqrt{n^2+n} + 3n}$$

$$= \lim_{n \rightarrow \infty} \frac{(\sqrt{n^2+n})^2 - (3n)^2}{\sqrt{n^2+n} + 3n}$$

$$= \lim_{n \rightarrow \infty} \frac{n^2+n-9n^2}{\sqrt{n^2+n} + 3n}$$

$$= \lim_{n \rightarrow \infty} \frac{-8n^2+n}{\sqrt{n^2+n} + 3n}$$

$$= \lim_{n \rightarrow \infty} \frac{-8 + \frac{1}{n}}{\sqrt{1 + \frac{1}{n}} + 3}$$

$$= \frac{-8 + 0}{\sqrt{1+0} + 3}$$

$$= \frac{-8}{4}$$

$$= -2$$

Ans:-

$$[As, \lim_{n \rightarrow \infty} \frac{1}{n} = 0]$$

$$(24) \lim_{n \rightarrow -\infty} \frac{\sqrt{1+4n^6}}{2-n^3}$$

$$= \lim_{t \rightarrow \infty} \frac{\sqrt{1+4(+)^6}}{2-(+)^3}$$

$$= \lim_{t \rightarrow \infty} \frac{\sqrt{1+4t^6}}{2+t^3}$$

$$= \lim_{t \rightarrow \infty} \frac{\frac{\sqrt{1+4t^6}}{(t^3)^2}}{\frac{2+t^3}{t^3}}$$

$$= \lim_{t \rightarrow \infty} \frac{\sqrt{\frac{1}{t^6} + 4}}{\frac{2}{t^3} + 1}$$

$$= \lim_{t \rightarrow \infty} \frac{\sqrt{\frac{1}{t^6} + 4}}{\frac{2}{t^3} + 1}$$

$$= \frac{2+0}{0+1}$$

$$= 2$$

$$(28) \lim_{n \rightarrow -\infty} \frac{\sqrt{(4n^2+3n)+2n}}{1}$$

$$= \lim_{t \rightarrow \infty} \frac{\sqrt{4(+)^2+3(+)+2(+)}}{1}$$

$$= \lim_{t \rightarrow \infty} \sqrt{4t^2+3t+2t}$$

$$= \lim_{t \rightarrow \infty} \frac{(\sqrt{4t^2+3t+2t}-2t)(\sqrt{4t^2+3t+2t}+2t)}{(\sqrt{4t^2+3t+2t}+2t)}$$

$$\left(\begin{array}{l} \text{let, } n \rightarrow -t \\ n \rightarrow -\infty \\ -t \rightarrow -\infty \\ t \rightarrow \infty \end{array} \right)$$

$$n \rightarrow -\infty$$

$$-t \rightarrow -\infty$$

$$t \rightarrow \infty$$

$$\left[\text{As, } \lim_{t \rightarrow \infty} \frac{1}{n^6} = 0 \right]$$

similar for val #

$$\left(\begin{array}{l} \text{let, } n \rightarrow -t \\ n \rightarrow -\infty \\ -t \rightarrow -\infty \\ t \rightarrow \infty \end{array} \right)$$

$$\begin{aligned}
 &= \lim_{t \rightarrow \infty} \frac{(\sqrt{4t^2 - 3t})^2 - (2t)^2}{\sqrt{4t^2 - 3t} + 2t} \\
 &= \lim_{t \rightarrow \infty} \frac{4t^2 - 3t - 4t^2}{\sqrt{4t^2 - 3t} + 2t} \\
 &= \lim_{t \rightarrow \infty} \frac{-3t}{\sqrt{\frac{4t^2 - 3t}{t^2}} + \frac{2t}{t}} \\
 &= \lim_{t \rightarrow \infty} \frac{-3}{\sqrt{4 - \frac{3}{t}} + 2} \\
 &= \frac{-3}{2+2} \\
 &= \frac{-3}{4}
 \end{aligned}$$

Ans:-

Law of limits:-

1. $\lim_{n \rightarrow \infty} \frac{1}{n} = 0$

2. $\lim_{n \rightarrow 0} \frac{\sin n}{n} = 1$

$$\lim_{n \rightarrow 0} \frac{\sin 3n}{n}$$

$$= \lim_{n \rightarrow 0} \frac{3 \sin 3n}{3n}$$

$$= 3 \lim_{n \rightarrow 0} \frac{\sin 3n}{3n} \quad \left[\text{As, } \lim_{n \rightarrow 0} \frac{\sin u}{u} \right]$$

$$= 3 \times 1$$

$$= 3$$

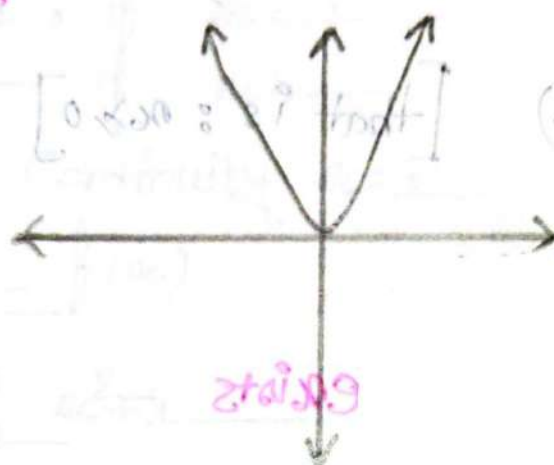
~~Continuity~~

Continuity:-

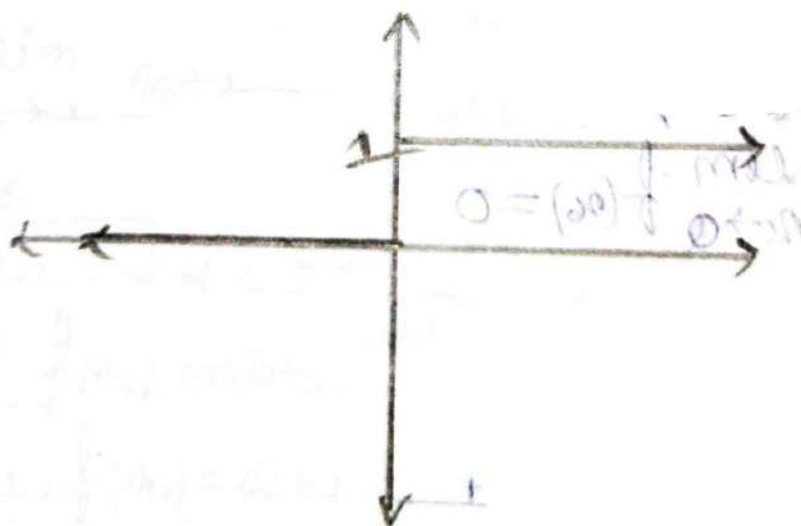
$$f(x) = \begin{cases} 1, & \text{if } x > 0 \\ 0, & \text{if } x < 0 \end{cases}$$

1. Continuous

$$y = x^2$$



2. Discontinuous function:-



$$x = 0, y = 1$$

$$x = 1, y = 1$$

$$x = -1, y = 0$$

$$x = -10, y = 0$$

Rules: —
non

1. Left hand limit is (L.H.L)

Ex: $\lim_{n \rightarrow a^-} f(n)$ [that is: $n < a$]

2. Right hand limit is (R.H.L)

Ex: $\lim_{n \rightarrow a^+} f(n)$ [that is: $n > a$]

if $L.H.L = R.H.L$

Then $\lim_{n \rightarrow a} f(n)$ exists

Q: $f(n) = \begin{cases} 1, & \text{if } n > 0 \\ 0, & \text{if } n < 0 \end{cases}$

$$\Rightarrow L.H.L = \lim_{n \rightarrow 0^-} f(n) \\ = \lim_{n \rightarrow 0^-} 0 = 0$$

$$R.H.L = \lim_{n \rightarrow 0^+} f(n) \\ = \lim_{n \rightarrow 0^+} 1 = 1$$

So, $L.H.L \neq R.H.L$

then $\lim_{n \rightarrow a} f(n)$ not exists

At $n = 0, f(n) = 1$

Points Law:

1. A function f is continuous at $x=a$

$$\text{if } \lim_{x \rightarrow a} f(x) = f(a)$$

Page 81828

$$(71) f(x) = \begin{cases} x^2 + 1, & \text{if } x < 1 \\ x + 1, & \text{if } x \geq 1 \end{cases}$$

Check the continuity $x=1$

$$\begin{aligned} \Rightarrow L.H.L. &= \lim_{x \rightarrow 1^-} f(x) \\ &= \lim_{x \rightarrow 1^-} x^2 + 1 \\ &= 2 \end{aligned}$$

$$\begin{aligned} R.H.L. &= \lim_{x \rightarrow 1^+} f(x) \\ &= \lim_{x \rightarrow 1^+} x + 1 \\ &= 2 \end{aligned}$$

$$\text{As, } L.H.L. = R.H.L. = 2$$

So, $\lim_{x \rightarrow 1} f(x)$ exists.

$$\text{At } x=1, f(x) = x + 1$$

$$\text{So, } f(1) = 1 + 1 = 2$$

$$\text{Now, } \lim_{x \rightarrow 1} f(x) = f(1) = 2$$

So, $f(x)$ is continuous at $x=1$

$$\textcircled{*} f(x) = \begin{cases} 2^x & ; \text{ if } x \leq 1 \\ 3-x & ; \text{ if } 1 < x \leq 4 \\ \sqrt{x} & ; \text{ if } x > 4 \end{cases}$$

check the continuity at $x=1$

$$\begin{aligned} \Rightarrow \text{L.H.L} &= \lim_{x \rightarrow 1^-} f(x) \\ &= \lim_{x \rightarrow 1^-} 2^x \\ &= 2 \end{aligned}$$

$$\begin{aligned} \text{R.H.L} &= \lim_{x \rightarrow 1^+} f(x) \\ &= \lim_{x \rightarrow 1^+} 3-x \\ &= 3-1 \\ &= 2 \end{aligned}$$

$$\text{As, L.H.L} = \text{R.H.L} = 2$$

$$\text{So, } \lim_{x \rightarrow 1} f(x) \text{ exist.}$$

$$\text{At, } x=1, f(x) = 2^x = 2^1 = 2$$

$$\text{So, } f(1) = 2^1 = 2$$

$$\text{Now, } \lim_{x \rightarrow 1} f(x) = f(1) = 2$$

So, $f(x)$ is continuous at $x=1$

27

$$(28) f(x) = \begin{cases} x^2, & \text{if } -1 \leq x \leq 0 \\ 2-3x, & \text{if } 0 \leq x \leq 1 \end{cases}$$

Check the continuity—

$$\begin{aligned} \Rightarrow \text{L.H.L} &= \lim_{x \rightarrow 0^-} f(x) \\ &= \lim_{x \rightarrow 0^-} x^2 \\ &= 0 \end{aligned}$$

$$\begin{aligned} \text{R.H.L} &= \lim_{x \rightarrow 0^+} f(x) \\ &= \lim_{x \rightarrow 0^+} 2-3x \\ &= 2 \end{aligned}$$

As, L.H.L $\neq$ R.H.L

So, $\lim_{x \rightarrow 0} f(x)$ not exist.

So, $f(x)$ is not Continuous at $x=0$.

$$\begin{aligned} \text{At, } x=0, f(x) &= x^2 \\ f(0) &= 0 \end{aligned}$$

$$(28) f(x) = \begin{cases} 2x+1, & \text{if } 0 \leq x < 1 \\ 4-2x, & \text{if } 1 \leq x \leq 2 \end{cases}$$

Check the continuity:—

$$\begin{aligned} \text{L.H.L} &= \lim_{x \rightarrow 1^-} f(x) \\ &= \lim_{x \rightarrow 1^-} 2x+1 \\ &= 2+1 \\ &= 3 \end{aligned}$$

$$\begin{aligned} \text{R.H.L} &= \lim_{x \rightarrow 1^+} f(x) \\ &= \lim_{x \rightarrow 1^+} 4-2x \\ &= 4-2 \times 1 \\ &= 2 \end{aligned}$$

$L.H.L \neq R.H.L$

- So $\lim_{n \rightarrow 1} f(n)$ does not exist

So, the limit does not exist.

So, $f(n)$ is not continuous at $n=1$

Ans:-

Differentiation:-

First Principle of Differentiation

$$\underline{1.} \quad \frac{d}{dx} (x) = 1$$

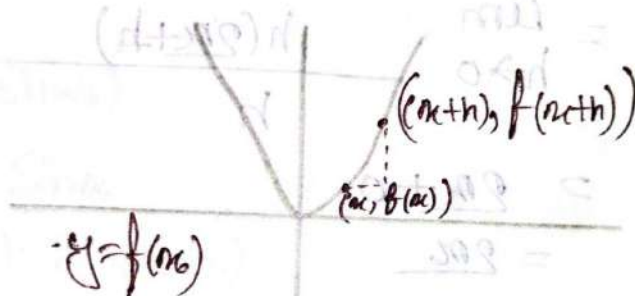
$$\underline{2.} \quad \frac{d}{dx} (x^2) = 2x$$

$$\underline{3.} \quad \frac{d}{dx} (x^3) = 3x^2$$

$$\underline{4.} \quad \frac{d}{dx} (\sin x) = \cos x$$

$$\underline{5.} \quad \frac{d}{dx} (\cos x) = -\sin x$$

Q:- Type 1



$$\frac{\Delta y}{\Delta x} = \frac{f(x+h) - f(x)}{x+h - x}$$

$$= \frac{f(x+h) - f(x)}{h}$$

$$\text{if } \lim_{h \rightarrow 0} \frac{\Delta y}{\Delta x} = \frac{dy}{dx}$$

$$\boxed{\frac{dy}{dx} = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}}$$

Q:- Using first principle of differentiation find $\frac{d}{dx}(x^2)$

$\Rightarrow$ Let, $f(x) = x^2$
 $f(x+h) = (x+h)^2$

We know,

$$\begin{aligned}\frac{d}{dx}(f(x)) &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \\&= \lim_{h \rightarrow 0} \frac{(x+h)^2 - x^2}{h} \\&= \lim_{h \rightarrow 0} \frac{x^2 + 2xh + h^2 - x^2}{h} \\&= \lim_{h \rightarrow 0} \frac{h(2x+h)}{h} \\&= 2x + 0 \\&= 2x\end{aligned}$$

Ans:-

Q:- $\frac{d}{dx}(x^2 + x)$

Let, $f(x) = x^2 + x$
 $f(x+h) = (x+h)^2 + (x+h)$

We know,

$$\begin{aligned}\frac{d}{dx}(f(x)) &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \\&= \lim_{h \rightarrow 0} \frac{(x+h)^2 + (x+h) - (x^2 + x)}{h}\end{aligned}$$

$$= \lim_{h \rightarrow 0} \frac{x^2 + 2xh + h^2 + x + h - x^2 - x}{h}$$

$$= \lim_{h \rightarrow 0} \frac{2xh + h^2 + h}{h}$$

$$= \lim_{h \rightarrow 0} \frac{h(2x + h + 1)}{h}$$

$$= \lim_{h \rightarrow 0} 2x + h + 1$$

$$= 2x + 0 + 1$$

$$= 2x + 1$$

Ans:—

Q:- $\frac{d}{dx} (\sin x)$

Let, $f(x) = \sin x$

$f(x+h) = \sin(x+h)$

We know,

$$\frac{d}{dx} (f(x)) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{\sin(x+h) - \sin x}{h}$$

$$= \lim_{h \rightarrow 0} \frac{\sin x \cosh + \cos x \sinh - \sin x}{h}$$

$$= \lim_{h \rightarrow 0} \frac{\sin x \cdot \cosh - \sin x + \cos x \cdot \sinh}{h}$$

$$= \lim_{h \rightarrow 0} \frac{\sin x (\cosh - 1) + \cos x \cdot \sinh}{h} =$$

$$= \sin \alpha \lim_{h \rightarrow 0} \frac{\cos h - 1}{h} + \cos \alpha \lim_{h \rightarrow 0} \frac{\sin h}{h}$$

$$= \sin \alpha \times 0 + \cos \alpha \times 1$$

$$= \cos \alpha$$

Ans:-

$$\textcircled{6} \lim_{\alpha \rightarrow 0} \frac{\cos \alpha - 1}{\alpha} = 0$$

$$\textcircled{7} \lim_{\alpha \rightarrow 0} \frac{\sin \alpha}{\alpha} = 1$$

Q:- $\frac{d}{d\alpha} \cos(\alpha)$

Let, $f(\alpha) = \cos(\alpha)$

$$f(\alpha+h) = \cos(\alpha+h)$$

We know,

$$\frac{d}{d\alpha} (f(\alpha)) = \lim_{h \rightarrow 0} \frac{f(\alpha+h) - f(\alpha)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{\cos(\alpha+h) - \cos \alpha}{h}$$

$$= \lim_{h \rightarrow 0} \frac{\cos \alpha \cdot \cos h - \sin \alpha \cdot \sin h - \cos \alpha}{h}$$

$$= \lim_{h \rightarrow 0} \frac{\cos \alpha (\cos h - 1) - \sin \alpha \cdot \sin h}{h}$$

=

$$\begin{aligned}
 &= \cos x \lim_{h \rightarrow 0} \frac{\cos(x+h) - 1}{h} - \sin x \lim_{h \rightarrow 0} \frac{\sin(x+h)}{h} \\
 &= \cos x \times 0 - \sin x \times 1 \\
 &= -\sin x
 \end{aligned}$$

Ans:-

Table of Differentiation

$$1. \frac{d}{dx}(x^n) = nx^{n-1}$$

$$2. \frac{d}{dx}(e^x) = e^x$$

$$3. \frac{d}{dx}(e^{mx}) = me^{mx}$$

$$4. \frac{d}{dx}(e^{-x}) = -e^{-x}$$

$$5. \frac{d}{dx}(e^{-mx}) = -me^{-mx}$$

$$6. \frac{d}{dx}(\sin x) = \cos x$$

$$7. \frac{d}{dx}(\cos x) = -\sin x$$

$$8. \frac{d}{dx}(uv) = u \frac{d}{dx}(v) + v \frac{d}{dx}(u)$$

$$9. \frac{d}{dx}\left(\frac{u}{v}\right) = \frac{v \frac{d}{dx}(u) - u \frac{d}{dx}(v)}{v^2}$$

$$10. \frac{d}{dx} \sqrt{x} = \frac{1}{2\sqrt{x}}$$

$$11. \frac{d}{dx} \ln(x) = \frac{1}{x}$$

$$12. \frac{d}{dx} \tan x = \sec^2 x$$

$$12. \frac{d}{dx} \tan x = \sec^2 x$$

Page: 188

③ $f(x) = (3x^2 - 5x)e^{2x}$

or $f(x) = 3x^2 e^{2x} - 5x e^{2x}$

① Differentiating both side with respects to x

$$\Rightarrow f'(x) = 3 \left[x^2 \frac{d}{dx} e^{2x} + e^{2x} \frac{d}{dx} x^2 \right] - 5 \left[x \frac{d}{dx} e^{2x} + e^{2x} \frac{d}{dx} x \right]$$

$$= 3 (x^2 e^{2x} + 2x e^{2x}) - 5 (x e^{2x} + e^{2x})$$

$$= 3x^2 e^{2x} + 6x e^{2x} - 5x e^{2x} - 5e^{2x}$$

$$= 3x^2 e^{2x} + x e^{2x} - 5e^{2x}$$

Ans:-

④ $g(x) = (x + 2\sqrt{x})e^{2x}$

$$\Rightarrow g'(x) = x e^{2x} + 2\sqrt{x} e^{2x}$$

$$= \left(x \frac{d}{dx} e^{2x} + e^{2x} \frac{d}{dx} x \right) + 2 \left(\sqrt{x} \frac{d}{dx} e^{2x} + e^{2x} \frac{d}{dx} \sqrt{x} \right)$$

$$= (x e^{2x} + e^{2x}) + 2 \left(\sqrt{x} e^{2x} + e^{2x} \frac{1}{2\sqrt{x}} \right)$$

$$= x e^{2x} + e^{2x} + 2\sqrt{x} e^{2x} + \frac{e^{2x}}{\sqrt{x}}$$

$$= e^{2x} \left(x + 1 + 2\sqrt{x} + \frac{1}{\sqrt{x}} \right)$$

Ans:-

$$\textcircled{*} \quad q(n) = \frac{1+2n}{3-4n}$$

$$\begin{aligned} \Rightarrow q'(n) &= \frac{(3-4n) \frac{d}{dn}(1+2n) - (1+2n) \frac{d}{dn}(3-4n)}{(3-4n)^2} \\ &= \frac{(3-4n)(0+2) - (1+2n)(0-4)}{(3-4n)^2} \\ &= \frac{6-8n+4+8n}{(3-4n)^2} \\ &= \frac{10}{(3-4n)^2} \end{aligned}$$

Ans: —

$$\textcircled{*} \quad y = \frac{n}{e^n}$$

$$\begin{aligned} \Rightarrow y' &= \frac{e^n \frac{d}{dn} n - n \frac{d}{dn} e^n}{(e^n)^2} \\ &= \frac{e^n - n e^n}{(e^n)^2} \\ &= \frac{e^n(1-n)}{(e^n)^2} \\ &= \frac{1-n}{e^n} \end{aligned}$$

Ans: —

11) $\frac{d}{dx} (\tan x) = \sec^2 x$

Page: 106

2) $f(x) = x \cdot \cos x + 2 \tan x$

$$\begin{aligned} \Rightarrow f'(x) &= x \cdot \frac{d}{dx} \cos x + 2 \cdot \frac{d}{dx} \tan x \\ &= x(-\sin x) + 2 \sec^2 x + \cos x \cdot \frac{d}{dx} x \\ &= -x \sin x + 2 \sec^2 x + \cos x \end{aligned}$$

Ans:

12) $y = \frac{\sin t}{1+t}$

$$\Rightarrow y' = \frac{(1+t) \left(\frac{d}{dt} \sin t \right) + \sin t \frac{d}{dt} (1+t)}{(1+t)^2} - \sin t \frac{d}{dt} (1+t)$$

$$= \frac{(1+t) (\cos t + \sin t) - \sin t (1+t)}{(1+t)^2}$$

$$= \frac{(1+t) (\cos t + \sin t) - \sin t (1+t)}{(1+t)^2}$$

Ans:

Exo 1 $f(x) = \cos(x^2)$

$$\begin{aligned} \Rightarrow f'(x) &= \frac{d}{dx} \cos(x^2) \\ &= -\sin(x^2) \times \frac{d}{dx} x^2 \\ &= -\sin(x^2) \times 2x \\ &= -2x \sin(x^2) \end{aligned}$$

Exo 2 $f(x) = \sin^2 x = (\sin x)^2$

$$\begin{aligned} f'(x) &= 2 \sin x \cdot \frac{d}{dx} \sin x \\ &= 2 \sin x \cdot \cos x \\ &= \sin 2x \end{aligned}$$

Exo 3 $f(x) = \cos^2 x = (\cos x)^2$

$$\begin{aligned} &= 2 \cos x \times \frac{d}{dx} \cos x \\ &= 2 \cos x \cdot (-\sin x) \\ &= -2 \sin x \cdot \cos x \end{aligned}$$

Page No. 2040

(14) $f(t) = t \sin \pi t$

$$\begin{aligned} f'(t) &= t \cdot \frac{d}{dt} \sin \pi t + \sin \pi t \cdot \frac{d}{dt} t \\ &= t \cdot \cos \pi t \cdot \frac{d}{dt} (\pi t) + \sin \pi t \cdot 1 \\ &= \pi t \cos \pi t + \sin \pi t \end{aligned}$$

Ans.

Implicit
Differentiation

Implicit Differentiation

Rules—

1. Differentiate an equation of $f(x, y) = c$ with respect to x .

2. Collect all $\frac{dy}{dx}$ in one side.

3. Solve for $\frac{dy}{dx}$

Ex: $x^2 + y^2 = 4$
 $f(x, y) = c$

Law

1. $\frac{d}{dx}(y) = \frac{dy}{dx}$

2. $\frac{d}{dx}(y^2) = 2y \frac{dy}{dx}$

3. $\frac{d}{dx}(y^3) = 3y^2 \frac{dy}{dx}$

4. $\frac{d}{dx}(xy) = x \frac{d}{dx}y + y \frac{dx}{dx}$
 $= x \frac{dy}{dx} + y \cdot 1$

5. $\frac{d}{dx}(xy^2) = 2xy \frac{dy}{dx} + y^2 \frac{dx}{dx}$
 $= 2xy \frac{dy}{dx} + y^2 \cdot 1$

Ex- $\frac{d}{dx}(x^2+y^2) = \frac{d}{dx}(4)$

$$\Rightarrow 2x \frac{dx}{dx} + 2y \frac{dy}{dx} = 0$$

$$\Rightarrow 2x \times 1 + 2y \frac{dy}{dx} = 0$$

$$\Rightarrow \frac{dy}{dx} = -\frac{2x}{2y} = -\frac{x}{y}$$

Ans:-

Page No:- 225:-

② $2x^2 + x + xy = 1$

$$\Rightarrow \frac{d}{dx}(2x^2 + x + xy) = \frac{d}{dx}(1)$$

$$\Rightarrow 2 \times 2x \frac{dx}{dx} + \frac{dx}{dx} + x \frac{dy}{dx} + y \frac{dx}{dx} = 0$$

$$\Rightarrow 4x + 1 + x \frac{dy}{dx} + y \times 1 = 0$$

$$\Rightarrow x \frac{dy}{dx} = -4x - 1 - y$$

$$\Rightarrow \frac{dy}{dx} = \frac{-(4x + y + 1)}{x}$$

Ans:-

③ $x^2 = 4xy + y^2$

$$\Rightarrow 2x - 4\left(x \frac{dy}{dx} + y\right) + 2y \frac{dy}{dx} = 0$$

$$\Rightarrow 2x - 4x \frac{dy}{dx} - 4y + 2y \frac{dy}{dx} = 0$$

$$\Rightarrow 2y \frac{dy}{dx} - 4x \frac{dy}{dx} = 4y - 2x$$

$$\Rightarrow \frac{dy}{dx}(2y - 4x) = 4y - 2x$$

$$\Rightarrow \frac{dy}{dx} = \frac{4x - 20}{2x - 12}$$

$$= \frac{2x - 10}{x - 6}$$

Ans

Parametric Differentiation

* if, $x = \cos t$
 $y = \sin t$

Find $\frac{dy}{dx}$

We can write

$$\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}} = \frac{\frac{d}{dt}(\sin t)}{\frac{d}{dt}(\cos t)} = \frac{\cos t}{-\sin t} = -\cot t$$

Ans

Page no: 655 (11-16) only

(11) Given, $x = t^2 + 1 \rightarrow$ find $\frac{dy}{dx}$
 $y = t^2 + t$

We can write

$$\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}} = \frac{2t + 1}{2t}$$

Ans

555

13

51

1

⑩ $m = t - \ln t$
 $f = t + \ln t$

$$\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}} = \frac{1 + \frac{1}{t}}{1 - \frac{1}{t}}$$

$$= \frac{\frac{t+1}{t}}{\frac{t-1}{t}}$$

$$= \frac{t+1}{t-1}$$

$$\frac{d}{dm}(\ln m) = \frac{1}{m}$$

⑪ $m = e^t$
 $f = te^t$

$$\Rightarrow \frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}} = \frac{\frac{d}{dt}(te^t)}{\frac{d}{dt}(e^t)}$$

$$= \frac{t \frac{d}{dt} e^t + e^t \frac{dt}{dt}}{e^t}$$

$$= \frac{-te^t + e^t \times 1}{e^t}$$

$$= \frac{e^t(1-t)}{e^t}$$

$$= \frac{(1-t)}{e^t \times e^t} = (1-t) \times e^{-2t}$$

Ans

Logarithmic Differentiations

Law —

$$\textcircled{1} \ln(mn) = \ln(m) + \ln(n)$$

$$\textcircled{2} \ln \frac{m}{n} = \ln m - \ln n$$

$$\textcircled{3} \ln(a^n) = n \ln a$$

Ex 1 — $y = m^x \rightarrow \ln y = \ln(m^x) = x \ln m$

$y = m^{\sin x} \rightarrow \ln y = \ln(m^{\sin x}) = \sin x \ln m$

$$y = \sqrt{\frac{2m+1}{3m-2}} \rightarrow \left\{ \frac{2m+1}{3m-2} \right\}^{\frac{1}{2}}$$

$$= \frac{1}{2} [\ln(2m+1) - \ln(3m-2)]$$

Page no. 223: —

$$\textcircled{Q10} y = (m^2+2)^2 (m^4+4)^4$$

$$\Rightarrow \ln(y) = \ln[(m^2+2)^2 (m^4+4)^4]$$

$$= \ln(m^2+2)^2 + \ln(m^4+4)^4$$

$$= 2 \ln(m^2+2) + 4 \ln(m^4+4)$$

$$\frac{d}{dm} \ln(y) = \frac{d}{dm} \{ 2 \ln(m^2+2) + 4 \ln(m^4+4) \}$$

$$\frac{1}{y} \frac{dy}{dm} = 2 \times \frac{1}{m^2+2} \times \frac{d}{dm}(m^2+2) + 4 \times \frac{1}{m^4+4} \times \frac{d}{dm}(m^4+4)$$

$$= 2 \times \frac{1}{m^2+2} \times 2m + 0 + 4 \times \frac{1}{m^4+4} \times 4m^3 + 0$$

$$= \frac{4m}{m^2+2} + \frac{16m^3}{m^4+4}$$

$$\frac{dy}{dm} = (m^2+2)^2 (m^4+4)^4 \left(\frac{4m}{m^2+2} + \frac{16m^3}{m^4+4} \right)$$

Ans —

43 $y = n^k$

$$\ln(y) = \ln(n^k)$$

$$\frac{1}{y} \frac{dy}{dn} = \frac{d}{dn} (n \ln n)$$

$$= n \times \frac{1}{n} + \ln n \times \frac{dn}{dn}$$

$$\frac{dy}{dn} = (1 + \ln n) \times y$$

$$= (1 + \ln n) \times n^k$$

Ans

* 44 $\sqrt{\frac{n-1}{n^4+1}} = y$

$$\Rightarrow y = \left(\frac{n-1}{n^4+1} \right)^{\frac{1}{2}}$$

$$\Rightarrow \ln y = \ln \left(\frac{n-1}{n^4+1} \right)^{\frac{1}{2}}$$

$$\Rightarrow \ln y = \frac{1}{2} [\ln(n-1) - \ln(n^4+1)]$$

$$\Rightarrow \frac{1}{y} \frac{dy}{dn} = \frac{1}{2} \left[\frac{1}{n-1} \times 1 - 0 - \frac{1}{n^4+1} \times 4n^3 \right]$$

$$= \frac{1}{2} \left[\frac{1}{n-1} - \frac{4n^3}{n^4+1} \right]$$

$$= \frac{1}{2} \left[\frac{1}{n-1} - \frac{4n^3}{n^4+1} \right] \times y$$

$$= \frac{1}{2} \left[\frac{1}{n-1} - \frac{4n^3}{n^4+1} \right] \times \left(\sqrt{\frac{n-1}{n^4+1}} \right)$$

(48) $y = (\sin x)^{\ln x}$

$\Rightarrow \ln y = \ln (\sin x)^{\ln x}$

$\Rightarrow \ln y = \ln x \cdot \ln (\sin x)$

$\Rightarrow \frac{1}{y} \frac{dy}{dx} = \ln x \times \frac{1}{\sin x} \cos x + \ln \sin x - x$
 $= \ln x \cot x + \frac{\ln \sin x}{x}$

$= y \times \left(\ln x \cot x + \frac{\ln \sin x}{x} \right)$
 $= (\sin x)^{\ln x} \times \left(\ln x \cot x + \frac{\ln \sin x}{x} \right)$
Ans

Differentiation of Inverse Trigonometric functions

Law:-

1. $\frac{d}{dx} (\sin^{-1} u) = \frac{1}{\sqrt{1-u^2}}$

2. $\frac{d}{dx} (\cos^{-1} u) = \frac{-1}{\sqrt{1-u^2}}$

3. $\frac{d}{dx} (\tan^{-1} u) = \frac{1}{1+u^2}$

4. $\tan^{-1} \rightarrow (\arctan u)$

Page: 214:—

Ex: (b) $f(x) = x \arctan \sqrt{x}$

$$\Rightarrow f(x) = x \tan^{-1} \sqrt{x}$$

$$\Rightarrow f'(x) = x \cdot \frac{1}{1+x^2} \times \frac{1}{2\sqrt{x}} + \tan^{-1} \sqrt{x} \times 1$$

$$= \frac{x}{(1+x^2) 2\sqrt{x}} + \tan^{-1} \sqrt{x}$$

$$= \frac{\sqrt{x} \times \sqrt{x}}{(1+x^2) 2\sqrt{x}} + \tan^{-1} \sqrt{x}$$

$$= \frac{\sqrt{x}}{2(1+x)} + \tan^{-1} \sqrt{x}$$

Ans

Ex: (c)

$$f(x) = \frac{1}{\sin^{-1} x}$$

$$f(x) = (\sin^{-1} x)^{-1}$$

$$\Rightarrow f'(x) = -1 \times \frac{1}{\sqrt{1-x^2}}$$

$$= \frac{-1}{\sqrt{1-x^2}}$$

Derivative of Hyperbolic functions

Laws:

- (1) $\frac{d}{dx} (\cosh x) = \sinh x$
- (2) $\frac{d}{dx} (\sinh x) = \cosh x$
- (3) $\frac{d}{dx} (\tanh x) = \operatorname{sech} x$

Page No: 264

(30) $f(x) = e^x \cosh x$

$$\Rightarrow f'(x) = e^x \sinh x + \cosh x e^x$$

$$= e^x (\sinh x + \cosh x)$$

Ans

(31) $f(x) = \tanh \sqrt{x}$

$$f'(x) = \operatorname{sech}^2 \sqrt{x} \times \frac{1}{2\sqrt{x}}$$

$$= \operatorname{sech}^2 \sqrt{x} / 2\sqrt{x}$$

Ans

(33) $h(x) = \sinh(x^2)$

$$h'(x) = \cosh(x^2) \times 2x$$

$$= 2x \cosh(x^2)$$

Ans

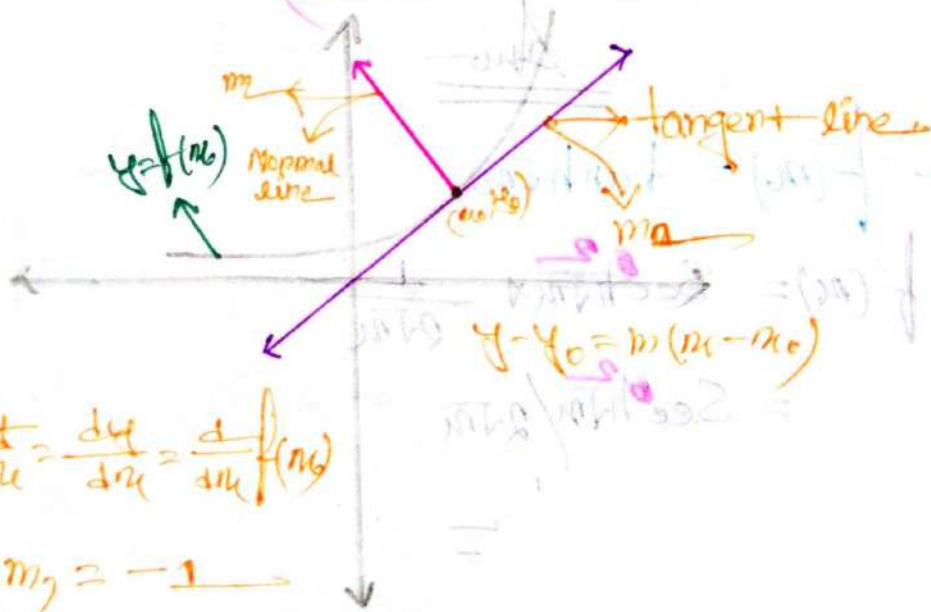
Q5) $G(t) = \sinh(\ln t)$

$\Rightarrow G'(t) = \cosh(\ln t) \times \frac{1}{t}$
 $= \frac{\cosh(\ln t)}{t}$

Ans: -

Application of Differentiation

Equation of tangent and Normal lines



$$m = \frac{dy}{dx} = \frac{dy}{dx} = \frac{d}{dx} f(x)$$

$$m_1 \times m_2 = -1$$

$$\text{So, } m_2 = -\frac{1}{m_1}$$

Exeg:

1. Equation of tangent line: $y - y_0 = m(x - x_0)$

where $m = \frac{dy}{dx}(x_0, y_0)$

2. Equation of Normal line: $y - y_0 = -\frac{1}{m}(x - x_0)$

62 ✓
25-5-20
8:10:38 AM
Page: 215

(27) Given, $x^2 - xy - y^2 = 1$ (2.17)

$\Rightarrow$ let, $(x_0, y_0) = (2, 1)$

and $x^2 - xy - y^2 = 1$

So, $\frac{d}{dx}(x^2 - xy - y^2) = \frac{d}{dx}(1)$

$\Rightarrow 2x - x \frac{dy}{dx} - y^2 - 2y \frac{dy}{dx} = 0$

$\Rightarrow 2x \frac{dy}{dx} + x \frac{dy}{dx} = 2x - y$

$\Rightarrow \frac{dy}{dx}(2x + x) = 2x - y$

$\Rightarrow \frac{dy}{dx} = \frac{2x - y}{2x + x}$

$\Rightarrow m = \frac{dy}{dx}(x_0, y_0) = \frac{2x_0 - y_0}{2x_0 + x_0}$

$= \frac{2 \times 2 - 1}{2 \times 2 + 2}$

$= \frac{3}{4}$

We know the equation of tangent line is:-

$y - y_0 = m(x - x_0)$

$y - 1 = \frac{3}{4}(x - 2)$

$y - 1 = \frac{3x}{4} - \frac{6}{4}$

$4y - 4 = 3x - 6$

$3x - 4y - 2 = 0$

And equation of normal line is

$$y - y_0 = -\frac{1}{m} (x - x_0)$$

$$\Rightarrow y - 1 = -\frac{4}{3} (x - 2)$$

$$\Rightarrow y - 1 = -\frac{4x}{3} + \frac{8}{3}$$

$$\Rightarrow 3y - 3 = -4x + 8$$

$$\Rightarrow \boxed{4x + 3y - 11 = 0}$$

Ans:-

Page: 216:-

Q. $x^2 + 2xy + 4y^2 = 12$, (2, 1)

Let, $\frac{d}{dx} (x^2 + 2xy + 4y^2) = \frac{d}{dx} (12)$

$$\Rightarrow 2x + 2\left(x \frac{dy}{dx} + y \cdot 1\right) + 4 \cdot 2y \frac{dy}{dx} = 0$$

$$\Rightarrow 2x + 2x \frac{dy}{dx} + 2y + 8y \frac{dy}{dx} = 0$$

$$\Rightarrow \frac{dy}{dx} (2x + 8y) = -2x - 2y$$

$$\Rightarrow \frac{dy}{dx} = \frac{-2x - 2y}{2x + 8y}$$

$$= \frac{2(-x - y)}{2(x + 4y)}$$

$$= \frac{-x - y}{x + 4y}$$

$$m = \frac{dy}{dx}(x_0, y_0) = \frac{-(x_0 + y_0)}{x_0 + 4y_0}$$

$$= \frac{-(2+1)}{2+4 \times 1}$$

$$= \frac{-3}{6}$$

$$= -\frac{1}{2}$$

we know,

the equation of tangent line is

$$y - y_0 = m(x - x_0)$$

$$\Rightarrow y - 1 = -\frac{1}{2}(x - 2)$$

$$\Rightarrow y - 1 = -\frac{1}{2}x + 1$$

$$\Rightarrow y - 1 = \frac{-x + 2}{2}$$

$$\Rightarrow 2y - 2 = -x + 2$$

$$\Rightarrow \boxed{x + 2y = 4}$$

the equation of normal line is

$$y - y_0 = -\frac{1}{m}(x - x_0)$$

$$y - 1 = -\frac{1}{-\frac{1}{2}}(x - 2)$$

$$y - 1 = 2(x - 2)$$

$$y - 1 = 2x - 4$$

$$\boxed{2x - y = 3}$$

Ans:-

Page: 658

③ Given, $x = t^3 + 1$
 $y = t^4 + t$; $t = -1$

$$x = (-1)^3 + 1 = -1 + 1 = 0$$

$$y = (-1)^4 + (-1) = 1 - 1 = 0$$

$$\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}} = \frac{\frac{d}{dt}(t^4 + t)}{\frac{d}{dt}(t^3 + 1)}$$

$$\frac{dy}{dx} = \frac{4t^3 + 1}{3t^2 + 0}$$

$$\frac{dy}{dx} = \frac{4(-1)^3 + 1}{3(-1)^2}$$

$$\frac{dy}{dx} = \frac{-4 + 1}{3}$$

$$m = \frac{dy}{dx}(x_0, y_0) = -1$$

we, know,

The equation of tangent line:

$$y - y_0 = m(x - x_0)$$

$$y - 0 = -1(x - 0)$$

$$y = -x$$

$$\boxed{x = -y}$$

Ans. —

The equation of normal line is:

$$y - y_0 = -\frac{1}{m} (x - x_0) \quad \text{--- (1)}$$

$$\Rightarrow y - 0 = -\frac{1}{-1} (x - 0) \quad \text{--- (2)}$$

$$\Rightarrow y = x$$

$$\Rightarrow \boxed{x = y}$$

Ans

④ $x_0 = \sqrt{t}$; $t = 4$
 $y = t^2 - t$

$$x_0 = \sqrt{4} = 2$$

$$y_0 = 4^2 - 4 = 16 - 4 = 12$$

$$\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}} = \frac{\frac{d}{dt}(t^2 - t)}{\frac{d}{dt}(\sqrt{t})}$$

$$= \frac{2t - 1}{\frac{1}{2\sqrt{t}}}$$

$$= \frac{2t - 1}{\frac{1}{2\sqrt{t}}}$$

$$= \frac{2 \times 4 - 1}{\frac{1}{2\sqrt{4}}}$$

$$= \frac{1}{2\sqrt{4}}$$

$$m = \frac{dy}{dx}(x_0, y_0) = 28$$

We know,

Equation of tangent line:

$$y - y_0 = m(x - x_0)$$

$$y - 12 = 28(x - 2)$$

$$y - 12 = 28x - 56$$

$$\boxed{28x - y = -44}$$

Equation of Normal line:-

$$(y - y_0) = -\frac{1}{m}(x - x_0)$$

$$(y - 12) = -\frac{1}{28}(x - 2)$$

$$\Rightarrow y - 12 = -\frac{x}{28} + \frac{2}{28}$$

$$\Rightarrow y - 12 = \frac{-x + 2}{28}$$

$$\Rightarrow 28y - 28 \times 12 = -x + 2$$

$$\Rightarrow 28y - 336 = -x + 2$$

$$\Rightarrow x + 28y = 338$$

Ans:-

① $x = t^2 - t$
 $y = t^2 + t + 1$; $(0, 3)$

$$\Rightarrow \begin{aligned} x &= t^2 - t \\ 0 &= t^2 - t \\ t^2 &= t \\ t &= 1 \end{aligned} \quad \left(\begin{aligned} y &= t^2 + t + 1 \\ 3 &= t^2 + t + 1 \\ t^2 + t - 2 &= 0 \\ t^2 + 2t - 1 - 2 &= 0 \\ t(t+2) - 1(t+2) &= 0 \\ t(t+2)(t-1) &= 0 \\ t &= 2, 1 \end{aligned} \right)$$

final common $t = 1$

$$\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}} = \frac{\frac{d}{dt}(t^2 + t + 1)}{\frac{d}{dt}(t^2 - t)}$$

$$\frac{dy}{dx} = \frac{2x+1}{2x-1}$$

$$m = \frac{dy}{dx}(x_0, y_0) = \frac{2(0)+1}{2(0)-1} = \frac{1}{-1} = -1$$

We know,

Equation of tangent line:—

$$y - y_0 = m(x - x_0)$$

$$\Rightarrow y - 3 = -1(x - 0)$$

$$\Rightarrow y - 3 = -x$$

$$\Rightarrow \boxed{x + y = 3}$$

Equation of Normal line:—

$$y - y_0 = -\frac{1}{m}(x - x_0)$$

$$\Rightarrow y - 3 = -\frac{1}{-1}(x - 0)$$

$$\Rightarrow y - 3 = x$$

$$\Rightarrow x - y + 3 = 0$$

$$\Rightarrow \boxed{x - y + 3 = 0}$$

Ans:—

Ex:- Determine the equation of the tangent and normal line to the curve given by $q(u) = 4u^3 + 12u^2 + 9u + 2$ at $u = -1$

$$\Rightarrow \text{Let } y = q(u) = 4(-1)^3 + 12(-1)^2 + 9(-1) + 2 = 1$$

$$q'(u) = 4 \times 3u^2 + 12 \times 2u + 9 \times 1 + 0$$

$$\begin{aligned} m = \frac{dy}{du}(u_0, y_0) &= 12u^2 + 24u + 9 \\ &= 12(-1)^2 + 24(-1) + 9 \\ &= 12 - 24 + 9 \\ &= -3 \end{aligned}$$

$$(u_0, y_0) = (-1, 1)$$

we know,

The equation of tangent line:-

$$y - y_0 = m(u - u_0)$$

$$y - 1 = m(u + 1)$$

$$y - 1 = -3u - 3$$

$$\boxed{3u + y = -2}$$

the equation of Normal line:-

$$y - y_0 = -\frac{1}{m}(u - u_0)$$

$$y - 1 = -\frac{1}{-3}(u + 1)$$

$$y - 1 = \frac{1}{3}u + \frac{1}{3}$$

$$3y - 3 = u + 1$$

$$\boxed{u - 3y = -4}$$

Ans.

Taylor and Maclaurin series:-

⇒ The function $f(x)$ has derivatives of all orders at $x=a$

≡ The Taylor series of $f(x)$ about $x=a$

$$f(x) = \sum_{k=0}^{\infty} \frac{f^{(k)}(a)}{k!} (x-a)^k = f(a) + \frac{f'(a)}{1!} (x-a) + \frac{f''(a)}{2!} (x-a)^2 + \dots$$

≡ The Maclaurin series of $f(x)$. [i.e. the Taylor series of $f(x)$ at $x=0$]

$$f(x) = \sum_{k=0}^{\infty} \frac{f^{(k)}(0)}{k!} x^k = f(0) + \frac{f'(0)}{1!} x + \frac{f''(0)}{2!} x^2 + \dots$$

Ex:- (p-74)

⑧ Find the Taylor series of the function $f(x) = \ln x$ about $x=1$ using first three non-zero terms.

$$\Rightarrow f(x) = \ln x \rightarrow f(1) = 0$$

$$\Rightarrow f'(x) = \frac{1}{x} \rightarrow f'(1) = \frac{1}{1} = 1$$

$$\Rightarrow f''(x) = \frac{x \cdot 0 - x \cdot 1}{x^2} \rightarrow f''(1) = \frac{-1}{1} = -1$$

$$\Rightarrow f'''(x) = -\frac{x^2 \cdot 0 - x \cdot 1}{(x^2)^2}$$

$$= -\frac{0 - x}{x^4} \rightarrow f'''(1) = -\left(\frac{-1}{1^4}\right) = -(-1) = 1$$

So, The Taylor series:-

$$\begin{aligned} f(x) &= f(1) + \frac{f'(1)}{1!} (x-1) + \frac{f''(1)}{2!} (x-1)^2 + \frac{f'''(1)}{3!} (x-1)^3 + \dots \\ &= 0 + \frac{1}{1} (x-1) + \frac{-1}{2} (x-1)^2 + \frac{1}{6} (x-1)^3 + \dots \end{aligned}$$

$$f(x) = (x-1) - \frac{1}{2}(x-1)^2 + \frac{1}{3}(x-1)^3 - \dots$$

13 Find the Maclaurin Series for the function $f(x) = \cos x$, using first three non-zero terms.

$$\Rightarrow f(x) = \cos x \rightarrow f(0) = \cos 0 = 1$$

$$\Rightarrow f'(x) = -\sin x \rightarrow f'(0) = -\sin 0 = 0$$

$$\Rightarrow f''(x) = -\cos x \rightarrow f''(0) = -\cos 0 = -1$$

$$\Rightarrow f'''(x) = -(-\sin x) \rightarrow f'''(0) = \sin 0 = 0$$

$$\Rightarrow f^{(4)}(x) = \cos x \rightarrow f^{(4)}(0) = \cos 0 = 1$$

So, the Maclaurin series:

$$\begin{aligned} f(x) &= f(0) + \frac{f'(0)}{1!}x + \frac{f''(0)}{2!}x^2 + \frac{f'''(0)}{3!}x^3 + \frac{f^{(4)}(0)}{4!}x^4 + \dots \\ &= 1 + 0 + \frac{-1}{2}x^2 + 0 + \frac{1}{24}x^4 + \dots \\ &= 1 - \frac{1}{2}x^2 + \frac{1}{24}x^4 - \dots \end{aligned}$$

Ans:-

14 $f(x) = xe^x$

$$\Rightarrow f(x) = xe^x \rightarrow f(0) = 0 \cdot e^0 = 0$$

$$\Rightarrow f'(x) = xe^x + e^x \rightarrow f'(0) = 0 + e^0 = 1$$

$$\Rightarrow f''(x) = xe^x + e^x + e^x \rightarrow f''(0) = 0 + 2e^0 = 2$$

$$\Rightarrow f'''(x) = xe^x + e^x + e^x + e^x \rightarrow f'''(0) = 0 + 3e^0 = 3$$

So, the Maclaurin Series:—

$$f(x) = f(0) + \frac{f'(0)}{1!}x + \frac{f''(0)}{2!}x^2 + \frac{f'''(0)}{3!}x^3 + \dots$$

$$= 0 + x + \frac{x^2}{2} + \frac{x^3}{6} + \dots$$

$$= x + \frac{x^2}{2} + \frac{x^3}{6} + \dots$$

Ans:—

(6) Find the Taylor series for the function $f(x) = \frac{1}{1+x}$ at the given value of $x_0 = 2$ using 1st non zero terms.

$$\Rightarrow f(x) = \frac{1}{1+x} \rightarrow f(2) = \frac{1}{1+2} = \frac{1}{3}$$

$$\Rightarrow f'(x) = \frac{1+x \cdot 0 - 1 \cdot (0+1)}{(1+x)^2}$$

$$= \frac{-1}{(1+x)^2}$$

$$= \frac{-1}{(1+2)^2}$$

$$= \frac{-1}{9}$$

$$f''(x) = \frac{(1+x)^2 \cdot 0 - 1 \cdot 2(1+x) \cdot 1}{\{(1+x)^2\}^2}$$

$$= \frac{-(-2-2x)}{(1+x)^4}$$

$$= \frac{+2+2x}{(1+x)^4}$$

$$= \frac{2}{27}$$

The Series of Taylor is:—

$$f(x) = f(2) + \frac{f'(2)}{1!}(x-2) + \frac{f''(2)}{2!}(x-2)^2 + \dots$$

$$= \frac{1}{3} + \frac{-\frac{1}{9}}{1}(x-2) + \frac{\frac{2}{27}}{2}(x-2)^2 + \dots$$

$$= \frac{1}{3} - \frac{1}{9}(x-2) + \frac{1}{27}(x-2)^2 + \dots$$

Ans:—

Different types of way:-

$$f(x) = \frac{1}{1+x} \Rightarrow (1+x)^{-1} \Rightarrow f(2) = (1+2)^{-1} = \frac{1}{3}$$

$$f'(x) = -1(1+x)^{-1-1} = -(1+x)^{-2} \Rightarrow f'(2) = -(1+2)^{-2} = -\frac{1}{9}$$

$$f''(x) = (-2(1+x)^{-2-1}) = 2(1+x)^{-3} \Rightarrow f''(2) = 2(1+2)^{-3} = \frac{2}{27}$$

So, the Taylor series:-

$$\begin{aligned} f(x) &= f(2) + \frac{f'(2)}{1!}(x-2) + \frac{f''(2)}{2!}(x-2)^2 + \dots \\ &= \frac{1}{3} + \frac{-\frac{1}{9}}{1}(x-2) + \frac{\frac{2}{27}}{2}(x-2)^2 + \dots \\ &= \frac{1}{3} - \frac{1}{9}(x-2) + \frac{1}{27}(x-2)^2 - \dots \end{aligned}$$

Ans:-

Q. Find the Maclaurin Series for the function
 $f(x) = (1-x)^{-2}$

$$\Rightarrow f(x) = (1-x)^{-2} \rightarrow f(0) = (1-0)^{-2} = 1$$

$$f'(x) = -2(1-x)^{-2-1}$$

$$= -2(1-x)^{-3} \rightarrow f'(0) = -2(1-0)^{-3} = -2$$

$$f''(x) = -2 \cdot -3(1-x)^{-3-1}$$

$$= 6(1-x)^{-4} \rightarrow f''(0) = 6(1-0)^{-4} = 6$$

So, the Maclaurin Series:-

$$f(x) = f(0) + \frac{f'(0)}{1!}x + \frac{f''(0)}{2!}x^2 + \dots$$

$$= 1 + \frac{-2}{1!}x + \frac{-6}{2!}x^2 + \dots$$

$$= 1 - 2x + \frac{-6}{2}x^2 + \dots$$

$$= 1 - 2x - 3x^2 + \dots$$

Ans:-

pg 771:

16 Find the Maclaurin Series for the function $f(x) = x \cos x$

$$\Rightarrow f(x) = x \cos x \longrightarrow f(0) = 0 \times \cos 0 = 0$$

$$f'(x) = x(-\sin x) + \cos x$$

$$= -x \sin x + \cos x \longrightarrow f'(0) = 0 + \cos 0 = 1$$

$$f''(x) = -(x \cos x + \sin x) + (-\sin x)$$

$$= -x \cos x - \sin x - \sin x$$

$$= -x \cos x - 2 \sin x$$

$$f'''(x) = -(x \cos x + \sin x) - 2(\cos x)$$

$$= -x \sin x - \cos x - 2 \cos x$$

$$= -x \sin x - 3 \cos x$$

$$\longrightarrow f'''(0) = 0 - 3 \cos 0 = -3$$

$$\begin{aligned}
 f''(u) &= u \cos u + \sin u - 3(-\sin u) \\
 &= u \cos u + \sin u + 3 \sin u \\
 &= u \cos u + 4 \sin u \quad \longrightarrow f''(0) = 0 + 4 \times \sin 0 = 0
 \end{aligned}$$

$$\begin{aligned}
 f'''(u) &= u(-\sin u) + \cos u + 4(\cos u) \\
 &= -u \sin u + \cos u + 4 \cos u \\
 &= -u \sin u + 5 \cos u \quad \longrightarrow f'''(0) = 0 + 5 \times \cos 0 = 5
 \end{aligned}$$

So, the Maclaurin Series —

$$\begin{aligned}
 f(u) &= f(0) + \frac{f'(0)}{1!} u + \frac{f''(0)}{2!} u^2 + \frac{f'''(0)}{3!} u^3 + \frac{f^{(4)}(0)}{4!} u^4 + \frac{f^{(5)}(0)}{5!} u^5 + \dots \\
 &= 0 + \frac{1}{1} u + \frac{0}{2} u^2 + \frac{-3}{6} u^3 + \frac{0}{24} u^4 + \frac{5}{120} u^5 + \dots \\
 &= u - \frac{1}{2} u^3 + \frac{5}{120} u^5 - \dots
 \end{aligned}$$

Ex 10 Find the Taylor Series for the function: —

① $f(u) = \cos 3u$, $u = a = \frac{\pi}{6}$

② $f(u) = \sin 5u$, $u = a = \frac{\pi}{3}$

⇒ ①

$$\begin{aligned}
 f(u) &= \cos 3u \quad \longrightarrow f\left(\frac{\pi}{6}\right) = \cos 3 \times \frac{\pi}{6} = \cos 3 \times \frac{180}{6} = 0 \\
 f'(u) &= -\sin 3u \quad \longrightarrow f'\left(\frac{\pi}{6}\right) = -\sin 3 \times \frac{180}{6} = -3
 \end{aligned}$$

$$f''(u) = -3(\cos 3u \times 3)$$

$$= -9 \cos 3u$$

$$f''(u) = -9 \cos 3u$$

$$= -9 \cos 3 \times \frac{180}{6} = 0$$

$$f'''(u) = -9(-\sin 3u \times 3)$$

$$= +27 \sin 3u$$

$$= 27 \sin 3u$$

$$f'''(u) = 27 \sin 3u \times \frac{180}{6} = 27$$

$$f^{iv}(u) = 27(\cos 3u \times 3)$$

$$= 81 \cos 3u$$

$$f^{iv}(u) = 81 \cos 3u \times \frac{180}{6} = 0$$

$$f^v(u) = 81(-\sin 3u \times 3)$$

$$= -243 \sin 3u$$

$$f^v(u) = -243 \sin 3 \times \frac{180}{6} = -243$$

So the Taylor Series—

$$f(u) = f\left(\frac{\pi}{6}\right) + \frac{f'\left(\frac{\pi}{6}\right)}{1!} \left(u - \frac{\pi}{6}\right) + \frac{f''\left(\frac{\pi}{6}\right)}{2!} \left(u - \frac{\pi}{6}\right)^2 + \frac{f'''\left(\frac{\pi}{6}\right)}{3!} \left(u - \frac{\pi}{6}\right)^3$$

$$+ \frac{f^{iv}\left(\frac{\pi}{6}\right)}{4!} \left(u - \frac{\pi}{6}\right)^4 + \frac{f^v\left(\frac{\pi}{6}\right)}{5!} \left(u - \frac{\pi}{6}\right)^5 + \dots$$

$$= 0 + \frac{-3}{1} \left(u - \frac{\pi}{6}\right) + 0 + \frac{27}{6} \left(u - \frac{\pi}{6}\right)^3 + 0 +$$

$$\frac{-243}{120} \left(u - \frac{\pi}{6}\right)^5 + \dots$$

$$= -3 \left(u - \frac{\pi}{6}\right) + \frac{9}{2} \left(u - \frac{\pi}{6}\right)^3 - \frac{81}{40} \left(u - \frac{\pi}{6}\right)^5 + \dots$$

Ans:-

② $\Rightarrow f(x) = \sin 5x \longrightarrow f\left(\frac{\pi}{3}\right) = \sin 5x \frac{150}{3} = -\frac{\sqrt{3}}{2}$

$$f'(x) = 5 \cos 5x$$

$$= 5 \cos 5x$$

$$\longrightarrow f\left(\frac{\pi}{3}\right) = 5 \cos 5x \frac{150}{3} = \frac{5}{2}$$

$$f''(x) = 5x(-\sin 5x \times 5)$$

$$= -25 \sin 5x$$

$$\longrightarrow f\left(\frac{\pi}{3}\right) = -25 \sin 5x \frac{150}{3} = -\frac{25\sqrt{3}}{2}$$

So, the Taylor Series—

$$f(x) = f\left(\frac{\pi}{3}\right) + \frac{f'\left(\frac{\pi}{3}\right)}{1!} \left(x - \frac{\pi}{3}\right) + \frac{f''\left(\frac{\pi}{3}\right)}{2!} \left(x - \frac{\pi}{3}\right)^2 + \dots$$

$$= -\frac{\sqrt{3}}{2} + \frac{5}{2} \left(x - \frac{\pi}{3}\right) + \frac{-25\sqrt{3}/2}{2} \left(x - \frac{\pi}{3}\right)^2 + \dots$$

$$= -\frac{\sqrt{3}}{2} + \frac{5}{2} \left(x - \frac{\pi}{3}\right) - \frac{25\sqrt{3}}{4} \left(x - \frac{\pi}{3}\right)^2 + \dots$$

Ans—

Short Questions—

1. Write Taylor Series of a function $f(x)$ at $x=a=2$

$\Rightarrow$ The Taylor Series is—

$$f(x) = f(2) + \frac{f'(2)}{1!} (x-2) + \frac{f''(2)}{2!} (x-2)^2 + \frac{f'''(2)}{3!} (x-2)^3 + \dots$$

Analysis of function:-

Analysis of functions:-

Stationary points

$\Rightarrow$ Gts a point of $f(x)$ where $f'(x) = 0$

Ex: what is the stationary point for the function of $f(x) = x^3 - 3x^2 - 9x + 4$

$\Rightarrow$ we know,

for stationary point $f'(x) = 0$

$$\Rightarrow 3x^2 - 6x - 9 = 0$$

$$\Rightarrow 3(x^2 - 2x - 3) = 0$$

$$\Rightarrow x^2 - 2x - 3 = 0$$

$$\Rightarrow x^2 - 3x + x - 3 = 0$$

$$\Rightarrow x(x-3) + 1(x-3) = 0$$

$$\Rightarrow (x+1)(x-3) = 0$$

$$\Rightarrow x = -1, 3$$

$$\text{for } f(-1) = (-1)^3 - 3(-1)^2 - 9(-1) + 4$$

$$= -1 - 3 + 9 + 4$$

$$= -1 - 3 + 9 + 4$$

$$= 9$$

$$\text{for } f(3) = (3)^3 - 3(3)^2 - 9(3) + 4$$

$$= 27 - 27 - 27 + 4$$

$$= -23$$

So Stationary points are $(-1, 9)$; $(3, -23)$

Ans:-

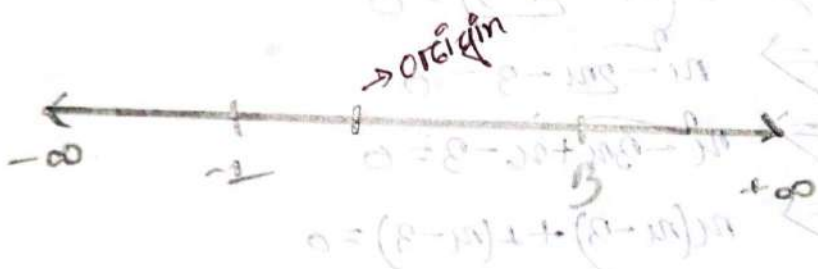
Increasing || Decreasing: (b)

⇒ On an Interval

① If $f'(x) > 0$, then $f(x)$ is increasing.

② If $f'(x) < 0$, then $f(x)$ is decreasing.

Ex: Find the intervals on which $f(x) = x^3 - 3x^2 - 9x + 4$ is increasing or decreasing.



$$f'(x) = 3x^2 - 6x - 9$$

$$\begin{aligned} f'(-2) &= 3 \times 4 - 12 - 9 \\ &= 12 - 12 - 9 \\ &= -9 < 0 \end{aligned}$$

$$f'(x) = 3x^2 - 6x - 9$$

$$\begin{aligned} f'(-2) &= 3(-2)^2 - 6(-2) - 9 \\ &= 12 + 12 - 9 \\ &= -9 + 24 \\ &= 15 > 0 \end{aligned}$$

$$\begin{aligned} f'(4) &= 3 \times (4)^2 - 6 \times 4 - 9 \\ &= 48 - 24 - 9 \\ &= 15 > 0 \end{aligned}$$

So, table of Analysis:—

Intervals	Sign in the Interval	Conclusion
$(-\infty, -1)$	$f'(x) = 3x^2 - 6x - 9 > 0$	Increasing
$(-1, 3)$	< 0	Decreasing
$(3, \infty)$	> 0	Increasing

Local Maximum & minimum (c)

Second Derivative test:—

⇒ At a Stationary point $x_0 = c$

i. If $f''(c) > 0$ then $f(x)$ has a local minimum at c

ii. If $f''(c) < 0$ then $f(x)$ has a local maximum at c

Ex: Find the local maximum and minimum value of $f(x) = x^3 - 3x^2 - 9x + 4$?

⇒ $f'(x) = 3x^2 - 6x - 9$
 $f''(x) = 6x - 6$

for $f''(-1) = 6 \times (-1) - 6$

$= -12 < 0 \rightarrow$ The maximum point

for $f''(3) = 3 \times 6 - 6$

$= 12 > 0 \rightarrow$ The minimum point

So, the value of maximum $f(-1) = x^3 - 3x^2 - 9x + 4$

$= 0 + 0 + 9 + 4$

$= (-1)^3 - 3 \times (-1)^2 - 9 \times (-1) + 4$

& the value of minimum $f(3) = x^3 - 3x^2 - 9x + 4$

$= (3)^3 - 3 \times (3)^2 - 9 \times 3 + 4$

$= -23$

(max, min) = $(9, -23)$

Ans.

Find the intervals of concavity
And the inflection point (d)

⇒ (i) Inflection point of $f(x)$ is where $f''(x) = 0$

(ii) Concavity—

⇒ for all x in an interval I

① If $f''(x) > 0$, so $f(x)$ is concave upwards on I

② If $f''(x) < 0$, then $f(x)$ is concave downwards on I

Ex— Find the intervals of concavity and inflection point for the function of $f(x) = x^3 - 3x^2 - 9x + 4$

⇒ $f(x) = x^3 - 3x^2 - 9x + 4$

$f'(x) = 3x^2 - 3 \times 2x - 9$

$f''(x) = 3 \times 2x - 6$

$= 6x - 6$

for inflection points—

$f''(x) = 0$

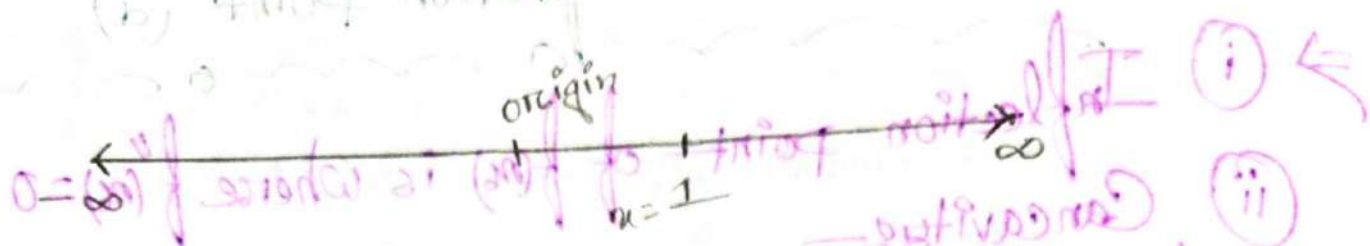
$6x - 6 = 0$

$x = 1$

$y = (1)^3 - 3(1)^2 - 9(1) + 4$
 $= -7$

So,
inflection point is $(1, -7)$

for Concavity:



Conditions

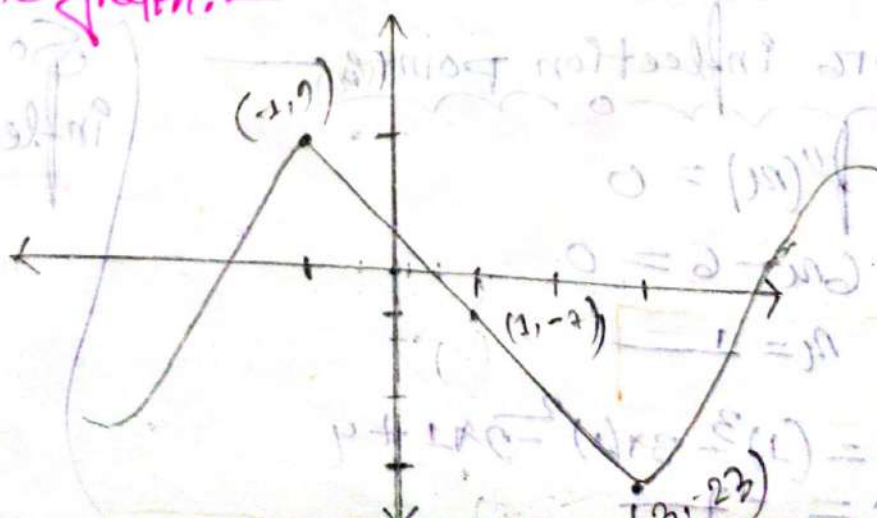
1. $(-\infty, 1) \rightarrow f''(x) = 6x - 6 = -6 < 0 \rightarrow \text{Concave down}$
2. $(1, \infty) \rightarrow f''(x) = 6x - 6 = 12 - 6 = 6 > 0 \rightarrow \text{Concave up}$

Sketch the graph

$\rightarrow$ for graph we need to know

1. Stationary point
2. Inflection point
3. Concavity

Ex: Use the information from parts (a-d) to sketch the graph:-



86

1. Stationary point = $(-1, 2)$ $(2, -2)$

2. Inflection point = $(1, -2)$

Summary of stationary points

1. Stationary point; $f'(x) = 0$

2. Increasing || decreasing;

(i) $f'(c) > 0$ & $\rightarrow$ increasing

(ii) $f'(c) < 0$ $\rightarrow$ decreasing

Here $c = x$ value of stationary point

3. Local Maximum || minimum;

(i) $f''(c) > 0$ $\rightarrow$ local minimum

(ii) $f''(c) < 0$ $\rightarrow$ local maximum

4. Inflection point: $f''(x) = 0$

5. Concavity;

(i) $f''(x) > 0$ $\rightarrow$ Concave up on I

(ii) $f''(x) < 0$ $\rightarrow$ Concave down on I

7:30:10 find the value of all previous type for the function of $f(x) = 2x^3 - 9x^2 + 12x - 3$

⇒ 1. Stationary points $f'(x) = 0$

$f(x) = 2x^3 - 9x^2 + 12x - 3$

⇒ $f'(x) = 3 \times 2x^2 - 9 \times 2x + 12$

⇒ $0 = 6x^2 - 18x + 12$

⇒ $6(x^2 - 3x + 2) = 0$

⇒ $x^2 - 3x + 2 = 0$

⇒ $x^2 - 2x - x + 2 = 0$

⇒ $(x-2)(x-1) = 0$

∴ $x = 2, 1$

$f(2) = 2 \times (2)^3 - 9 \times (2)^2 + 12 \times 2 - 3$

$= 16 - 36 + 24 - 3$
 $= 1$

$f(1) = 2 \times (1)^3 - 9 \times (1)^2 + 12 \times 1 - 3$

$= 2 - 9 + 12 - 3$

$= 2$

So, Stationary point are $(2, 1), (1, 2)$

2. Increasing 1. Decreasing
 ① $f'(x) > 0 \rightarrow$ in
 ② $f'(x) < 0 \rightarrow$ dec

$$f(x) = 2x^3 - 9x^2 + 12x - 3$$

$$f'(x) = 6x^2 - 18x + 12$$

$$f'(x) = 0$$

$$6x^2 - 18x + 12 = 0$$

$$x = (1, 2)$$



Intervals:-

1. $(-\infty, 1) \rightarrow f'(0) = 6 \times 0^2 - 18 \times 0 + 12 = 12 > 0 \rightarrow$ increasing

2. $(1, 2) \rightarrow f'(1.5) = (1.5)^2 \times 6 - 18 \times (1.5) + 12 = -1.5 < 0 \rightarrow$ decreasing

3. $(2, \infty) \rightarrow f'(3) = 5^2 \times 6 - 18 \times 5 + 12 = 72 > 0 \rightarrow$ increasing

Table of Analysis:-

Intervals	Sign in the intervals $f'(x) = 6x^2 - 18x + 12$	Conclusion
$(-\infty, 1)$	> 0	increasing
$(1, 2)$	< 0	decreasing
$(2, \infty)$	> 0	increasing

2. Local Max / Mini: ① $f''(c) < 0$ local max ② $f''(c) > 0$ local mini $f'(c) = 0$ value of stationary point

$$\Rightarrow f(x) = 2x^3 - 9x^2 + 12x - 3$$

$$\Rightarrow f'(x) = 6x^2 - 18x + 12$$

$$\Rightarrow f''(x) = 12x - 18$$

$$x = (1, 2)$$

$$f''(1) = 12 \times 1 - 18 = -6 < 0 \text{ have a max value of this point}$$

$$f''(2) = 12 \times 2 - 18 = 6 > 0 \text{ have a mini value u u u}$$

So, the value of max. is

$$\begin{aligned} f(1) &= 2 \times (1)^3 - 9 \times (1)^2 + 12 \times 1 - 3 \\ &= 2 - 9 + 12 - 3 \\ &= 2 \end{aligned}$$

326079-101

the value of mini is

$$\begin{aligned} f(2) &= 2 \times (2)^3 - 9 \times 2^2 + 12 \times 2 - 3 \\ &= 16 - 36 + 24 - 3 \\ &= 1 \end{aligned}$$

interval	$0 <$	$(1, \infty)$
interval	$0 >$	$(\infty, 1)$
interval	$0 <$	$(1, \infty)$

4. Inflection point: ① $f''(n_6) = 0$

$$f(n_6) = 2n^3 - 9n^2 + 12n - 3$$

$$f'(n) = 6n^2 - 18n + 12$$

$$f''(n) = 12n - 18$$

$$0 = 12n - 18$$

$$n_6 = \frac{18}{12} = \frac{3}{2}$$

$$f(n_6) = 2 \times \left(\frac{3}{2}\right)^3 - 9 \times \left(\frac{3}{2}\right)^2 + 12 \times \left(\frac{3}{2}\right) - 3$$

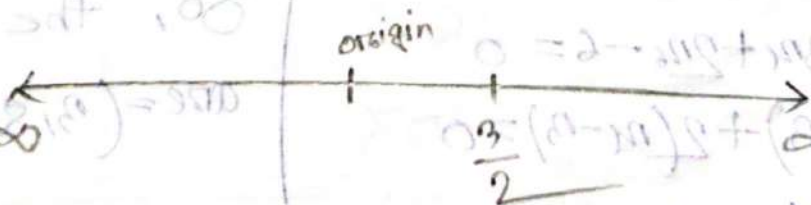
$$= \frac{3}{2}$$

So, inflection point is $\left(\frac{3}{2}, \frac{3}{2}\right)$

5. Concavity: ① $f''(n) > 0 \rightarrow$ concave up
② $f''(n) < 0 \rightarrow$ concave down

$$f''(n) = 12n - 18$$

$$n_6 = \frac{3}{2}$$

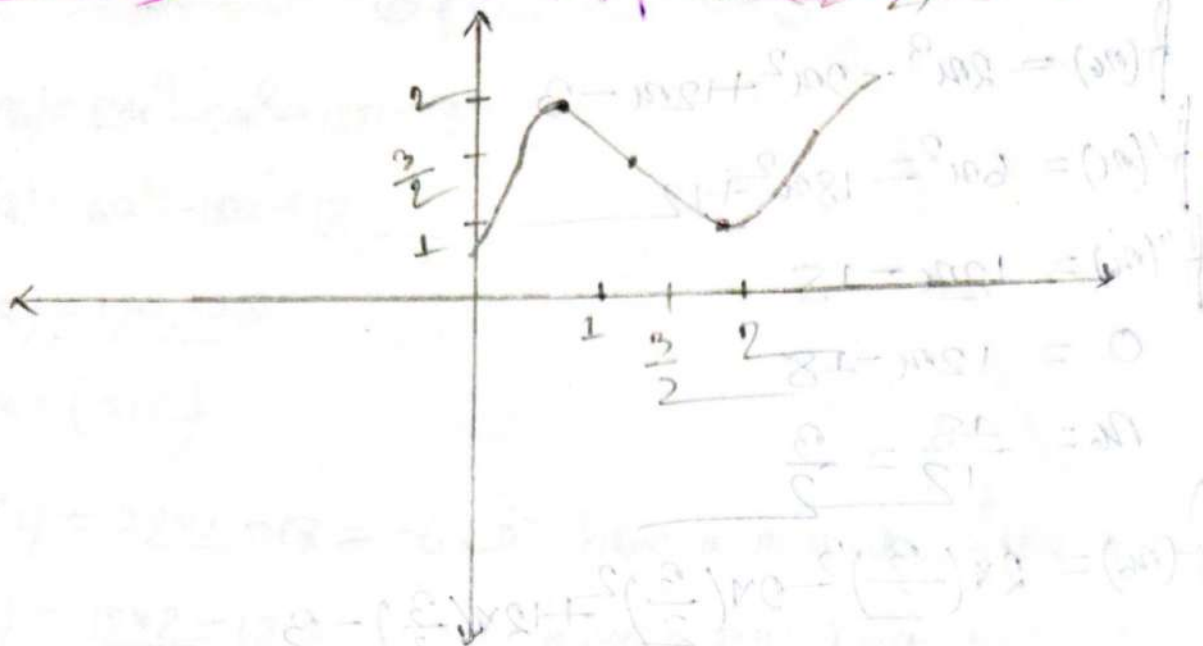


Intervals:

$$\left(-\infty, \frac{3}{2}\right) \rightarrow f''(0) = 12 \times 0 - 18 = -18 < 0 \rightarrow \text{concave down}$$

$$\left(\frac{3}{2}, \infty\right) \rightarrow f''(2) = 12 \times 2 - 18 = 6 > 0 \rightarrow \text{concave up}$$

6. Graph:-



Page: 302

Q38 $f(x) = 36x + 3x^2 - 2x^3$

1. Stationary points $f'(x) = 0$

$$f(x) = 36x + 3x^2 - 2x^3$$

$$\Rightarrow f'(x) = 36 + 6x - 6x^2$$

$$\Rightarrow 36 + 6x - 6x^2 = 0$$

$$\Rightarrow 6(6 + x - x^2) = 0$$

$$\Rightarrow x^2 - x - 6 = 0$$

$$\Rightarrow x^2 - 3x + 2x - 6 = 0$$

$$\Rightarrow x(x-3) + 2(x-3) = 0$$

$$\Rightarrow (x-3)(x+2) = 0$$

$$\Rightarrow x = 3, -2$$

$$\text{For } f(3) = 36 \times 3 + 3 \times (3)^2 - 2 \times (3)^3 = 81$$

$$f(-2) = 36 \times (-2) + 3 \times (-2)^2 - 2 \times (-2)^3 = -44$$

So, the stationary points are $(3, 81)$; $(-2, -44)$

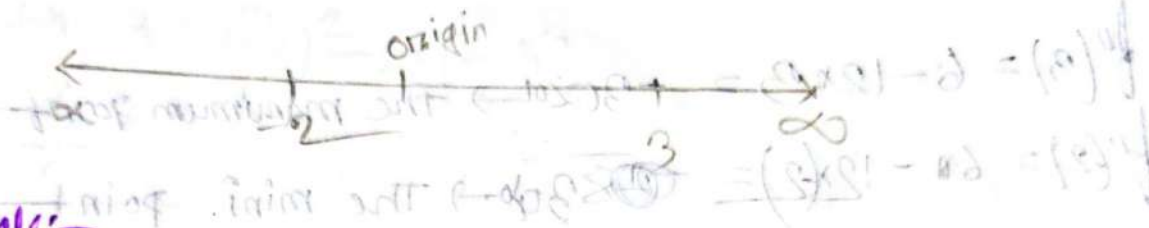
2. increasing || Decreasing : ① $f'(x) > 0 \rightarrow \text{in}$ ② $f'(x) < 0 \rightarrow \text{dec}$

$$f(x) = 36x + 3x^2 - 2x^3$$

$$f'(x) = 36 + 6x - 6x^2$$

$$f'(x) = 0$$

$$x = (3, 2)$$



Intervals:-

- ① $(-\infty, -2) \rightarrow f'(-3) = 36 + 6(-3) - 6(-3)^2 = -36 < 0 \rightarrow \text{Decreasing}$
- ② $(-2, 3) \rightarrow f'(2.5) = 36 + 6(2.5) - 6(2.5)^2 = 13.5 > 0 \rightarrow \text{Increasing}$
- ③ $(3, \infty) \rightarrow f'(5) = 36 + 6(5) - 6(5)^2 = -84 < 0 \rightarrow \text{Decreasing}$

Intervals	Sign in Intervals $f'(x) = 36 + 6x - 6x^2$	Conclusion
$(-\infty, -2)$	< 0	Decreasing
$(-2, 3)$	> 0	Increasing
$(3, \infty)$	< 0	Decreasing

③ Local max/min: (i) $f'(c) < 0 \rightarrow$ local max
 (ii) $f'(c) > 0 \rightarrow$ local min

$$f'(x) = 36 + 6x - 6x^2$$

$$x = (3, -2)$$

$$f''(x) = 0 + 6 - 12x$$

$$f''(3) = 6 - 12 \times 3 = -30 < 0 \rightarrow \text{the maximum point}$$

$$f''(-2) = 6 - 12 \times (-2) = 30 > 0 \rightarrow \text{The mini point}$$

So, the value of maximum is

$$f(3) = 36 \times 3 + 3 \times (3)^2 - 2 \times (3)^3$$

$$= 81$$

the value of minimum is

$$f(-2) = 36 \times (-2) + 3 \times (-2)^2 - 2 \times (-2)^3$$

$$= -44$$

4) Inflection points $f''(x) = 0$

$$f''(x) = 6 - 12x$$

$$0 = 6 - 12x$$

$$12x = 6$$

$$x = \frac{1}{2}$$

$$f(x) = 36x + 6x^2 = 2x^3$$

$$= 36 \times \frac{1}{2} + 6 \left(\frac{1}{2} \right)^2 = 2 \left(\frac{1}{2} \right)^3$$

$$= \frac{37}{2}$$

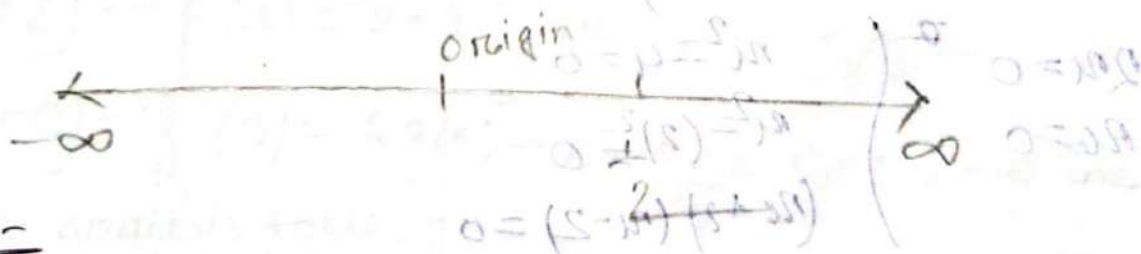
So, Inflection point : $\left(\frac{1}{2}, \frac{37}{2} \right)$

5) Concavity ① $f''(x) > 0 \rightarrow$ Concave up

② $f''(x) < 0 \rightarrow$ Concave down

$$f''(x) = 6 - 12x$$

$$x = \frac{1}{2}$$

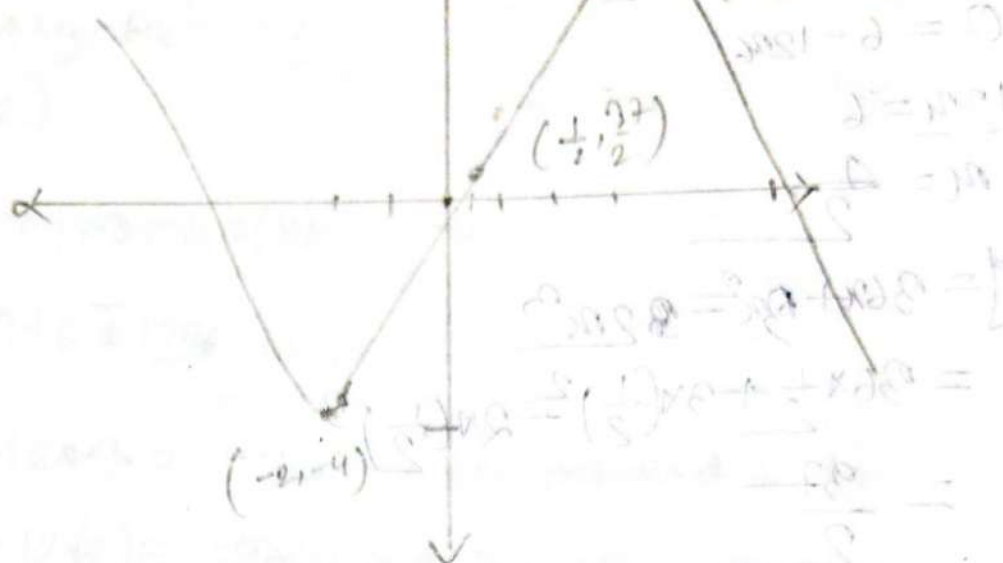


Intervals:-

$$(-\infty, \frac{1}{2}) \rightarrow f''(0) = 6 - 12 \times 0 = 6 > 0 \rightarrow \text{Concave up}$$

$$(\frac{1}{2}, \infty) \rightarrow f''(2) = 6 - 12 \times 2 = -18 < 0 \rightarrow \text{Concave down}$$

Graphs



Q1) $f(x) = \frac{1}{2}x^4 - 4x^2 + 3$

① Stationary points: $f'(x) = 0$

$$f(x) = \frac{1}{2}x^4 - 4x^2 + 3$$

$$f'(x) = \frac{1}{2} \times 4x^3 - 8x + 0$$

$$0 = 2x^3 - 8x$$

$$= 2x(x^2 - 4)$$

$$\begin{aligned} 2x &= 0 \\ x &= 0 \end{aligned} \quad \left\{ \begin{aligned} x^2 - 4 &= 0 \\ x^2 - (2)^2 &= 0 \\ (x+2)(x-2) &= 0 \\ x &= 2, -2 \end{aligned} \right.$$

for concave up $\rightarrow 0 < f''(x) \rightarrow 0 \times 51 - 2 = 0 \rightarrow f''(0) = 0 \rightarrow f''(0) = 0$

concave down $\rightarrow 0 > f''(x) \rightarrow 0 \times 51 - 2 = -2 < 0 \rightarrow f''(0) = -2 < 0$

$$f(2) = \frac{1}{2}(2)^4 - 4 \times (2)^2 + 3 = -5 \quad f(0) = 0 - 0 + 3 = 3$$

$$f(-2) = \frac{1}{2}(-2)^4 - 4 \times (-2)^2 + 3 = -5$$

Stationary point are $(2, -5)$, $(-2, -5)$, $(0, 3)$

Ans:-

② Increasing & decreasing: (i) $f'(x) < 0$ dec
(ii) $f'(x) > 0$ inc

$$f'(x) = 2x^3 - 8x$$

$$x = 0, 2, -2$$

Intervals:

1. $(-\infty, -2) \rightarrow f'(-3) = 2 \times (-3)^3 - 8 \times (-3) = -30 < 0 \rightarrow \text{dec}$

2. $(-2, 0) \rightarrow f'(-1) = 2 \times (-1)^3 - 8 \times (-1) = 6 > 0 \rightarrow \text{inc}$

3. $(0, 2) \rightarrow f'(1) = 2 \times 1^3 - 8 \times 1 = -6 < 0 \rightarrow \text{dec}$

4. $(2, \infty) \rightarrow f'(3) = 2 \times (3)^3 - 8 \times (3) = 30 > 0 \rightarrow \text{inc}$

So, the analysis table:-

Intervals	Sign of Intervals $f'(x) = 2x^3 - 8x$	Conclusion
$(-\infty, -2)$	< 0	Decreasing
$(-2, 0)$	> 0	Increasing
$(0, 2)$	< 0	Decreasing
$(2, \infty)$	> 0	Increasing

3. Max / mini: ① $f''(c) > 0 \rightarrow$ mini value
 ② $f''(c) < 0 \rightarrow$ max value

$$f(x) = \frac{1}{2}x^4 - 4x^2 + 3$$

$$f'(x) = 2x^3 - 8x + 0$$

$$f''(x) = 6x^2 - 8$$

$$x = 0, 2, -2$$

$$f''(0) = 6 \times (0)^2 - 8 = -8 < 0 \rightarrow \text{maximum point}$$

$$\left. \begin{aligned} f''(2) &= 6 \times (2)^2 - 8 = 16 > 0 \\ f''(-2) &= 6 \times (-2)^2 - 8 = 16 > 0 \end{aligned} \right\} \rightarrow \text{minimum point}$$

So, the value of maximum is

$$f(0) = \frac{1}{2}(0)^4 - 4 \times (0)^2 + 3 = 3$$

the value of minimum is

$$f(2) = \frac{1}{2}(2)^4 - 4 \times (2)^2 + 3 = 2 - 8$$

Ans:-

4. Inflection Point: ① $f''(u) = 0$

$$f(u) = \frac{1}{2}u^4 - 4u^2 + 3$$

$$f'(u) = 2u^3 - 8u + 0$$

$$f''(u) = 6u^2 - 8$$

$$0 = 6u^2 - 8$$

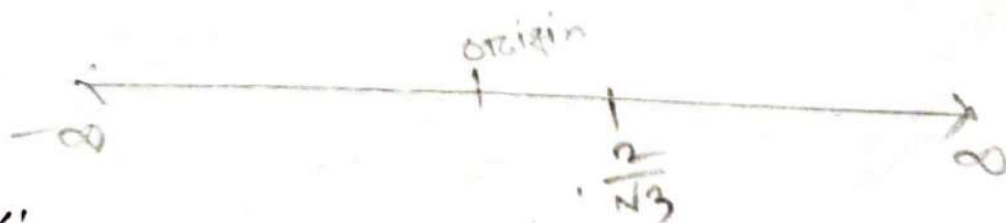
$$6u^2 = 8$$

$$u = \sqrt{\frac{8}{6}} = \frac{2\sqrt{3}}{3}$$

$$f(u) = y = \frac{1}{2}\left(\frac{2\sqrt{3}}{3}\right)^4 - 4\left(\frac{2\sqrt{3}}{3}\right)^2 + 3$$
$$= -\frac{13}{9}$$

So, Inflection point is $= \left(\frac{2}{\sqrt{3}}, -\frac{13}{9}\right)$

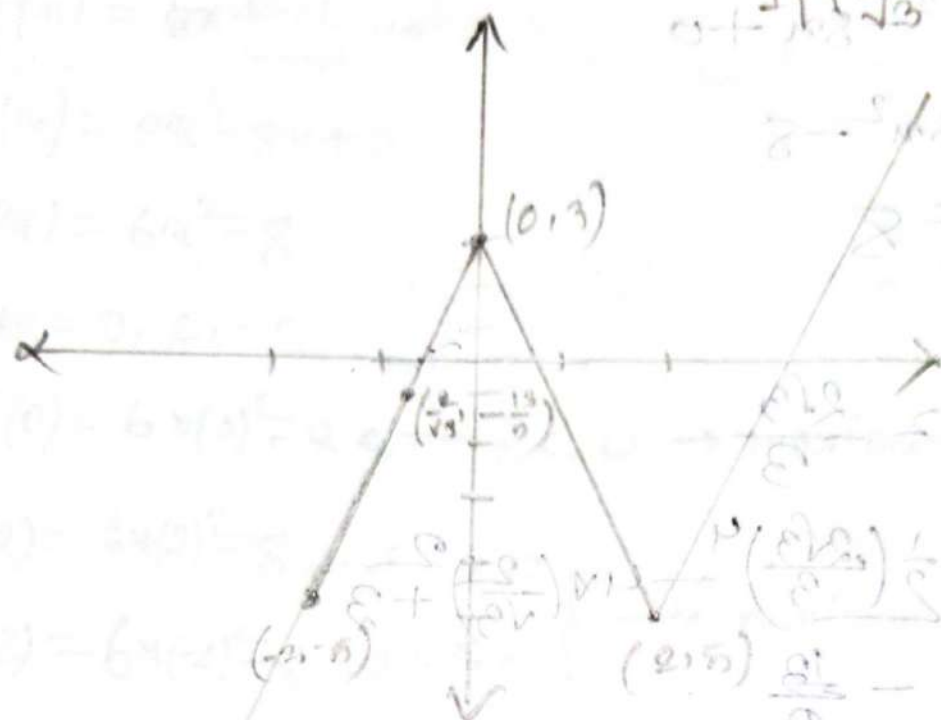
Concavity: ① $f''(u) > 0 \rightarrow$ concave up
② $f''(u) < 0 \rightarrow$ concave down



Intervals:

1. $(-\infty, \frac{2}{\sqrt{3}}) \rightarrow f''(0) = 6 \times 0^2 - 8 = -8 < 0 \rightarrow$ concave down
2. $(\frac{2}{\sqrt{3}}, \infty) \rightarrow f''(2) = 6 \times (2)^2 - 8 = 16 > 0 \rightarrow$ concave up

Graph:



① Derivative mit $f'(x) = 0$

$$SP(2, 5) \quad (2, 5) \quad (0, 3)$$

$$IP\left(\frac{2}{\sqrt{3}}, \frac{10}{3}\right)$$

$$2 - \frac{1}{2} = (1, 2)$$

$$2 - \frac{1}{2} = 0$$

$$2 - \frac{1}{2} = 0$$

$$2 - \frac{1}{2} = 0$$

$$f'(x) = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$$

$$\frac{1}{2} - \frac{1}{2} = 0$$

② Derivative mit $f'(x) = 0$

① Concavity:
 (i) $f''(x) > 0 \rightarrow$ concave up
 (ii) $f''(x) < 0 \rightarrow$ concave down

Derivative

$$f'(x) = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$$

$$f'(x) = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$$

✓ Hospital Rule:

103

pg 311

$$(51) \lim_{n \rightarrow 1} \left(\frac{n_0}{n-1} - \frac{1}{\ln n_0} \right)$$

$$\Rightarrow \lim_{n \rightarrow 1} \left(\frac{n \times \ln n - (n-1)}{(n-1)(\ln n)} \right)$$

$$\Rightarrow \lim_{n \rightarrow 1} \frac{n \times \frac{1}{n} + \ln n - 1}{(n-1) \times \frac{1}{n} + \ln n (1)}$$

$$\Rightarrow \lim_{n \rightarrow 1} \frac{1 - 1 + \ln n_0}{1 - \frac{1}{n_0} + \ln n_0}$$

$$\Rightarrow \lim_{n \rightarrow 1} \frac{\frac{1}{n_0}}{0 - \frac{1}{n^2} + \frac{1}{n_0}}$$

$$\Rightarrow \lim_{n \rightarrow 1} \frac{\frac{1}{n}}{\frac{1}{n^2} + \frac{1}{n}}$$

$$\Rightarrow \frac{\frac{1}{1}}{\frac{1}{1^2} + \frac{1}{1}}$$

$$= \frac{1}{2}$$

Ans:-

(15) $\lim_{t \rightarrow 0} \frac{2^t - 1}{\sin t}$

$\Rightarrow \lim_{t \rightarrow 0} \frac{2^t - 1}{\cos t}$

$\Rightarrow \frac{2^0 - 1}{\cos 0}$

$= 2$

(16) $\lim_{x \rightarrow \infty} x^3 - x^2$

$\Rightarrow \lim_{x \rightarrow \infty} \frac{x^3}{e^{x^2}}$

$\Rightarrow \lim_{x \rightarrow \infty} \frac{3x^2}{2xe^{x^2}}$

$\Rightarrow \lim_{x \rightarrow \infty} \frac{3x}{2e^{x^2}}$

$\Rightarrow \lim_{x \rightarrow \infty} \frac{3}{4xe^{x^2}}$

$\Rightarrow \frac{3}{\infty \times \infty}$

$= 0$

$\left(\frac{1}{\cos x} - \frac{\sin x}{1 - \cos x} \right) \cdot \sin x$

law of

Something
Infinity $= 0$

(17) $\lim_{x \rightarrow \infty} \frac{100}{\infty} = 0$

$\frac{1}{\infty} + \frac{1}{\infty} = 0$

$\frac{1}{\infty} + \frac{1}{\infty} = 0$

$\frac{1}{\infty} + \frac{1}{\infty} = 0$

$\frac{1}{\infty} + \frac{1}{\infty} = 0$

$\frac{1}{\infty} + \frac{1}{\infty} = 0$

$\frac{1}{\infty} + \frac{1}{\infty} = 0$

$\frac{1}{\infty} + \frac{1}{\infty} = 0$

Ans:

$\Rightarrow \lim_{n \rightarrow \infty} \ln n^2 = \underline{\hspace{2cm}} \quad (1)$

NOLON

$\Rightarrow$ Lcm $n \rightarrow \infty$ no Lcm

$$\Rightarrow \lim_{n \rightarrow 0^+} \frac{\ln nx}{\frac{1}{nx}}$$

$$\Rightarrow \lim_{n \rightarrow 0^+} \frac{\frac{1}{n}}{-\frac{1}{n^2}}$$

$$\Rightarrow \lim_{n \rightarrow 0^+} \frac{1}{n} - \frac{n^2}{1}$$

$$\Rightarrow \lim_{n \rightarrow 0^+} -\infty$$

11 0

So, from ① $\Rightarrow$

$$\lim_{n \rightarrow 0^+} (n^n) = \lim_{n \rightarrow 0^+} e^0 = 1$$

$\frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$

$\frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$

Law:-

$$\lim_{n \rightarrow \infty} e^n = \infty$$

207

$\frac{1}{2}$
 $\frac{1}{2}$
 $\frac{1}{2}$

$$\left(\frac{1}{2} \right)$$

(10/10)

$$\# \lim_{t \rightarrow 0} \frac{e^t - 1}{\sin t}$$

$$\Rightarrow \lim_{t \rightarrow 0} \frac{e^t - 0}{\text{conf}}$$

$$= \frac{2 \times e - 0}{\text{conf}}$$

$$= 2$$

$$\# \lim_{n \rightarrow \infty} \frac{\ln n}{\sqrt{n}}$$

$$\Rightarrow \lim_{n \rightarrow \infty} \frac{1/n}{\frac{1}{2\sqrt{n}}}$$

$$\Rightarrow \lim_{n \rightarrow \infty} \left(\frac{1}{n} \times \frac{2\sqrt{n}}{1} \right)$$

$$\Rightarrow \lim_{n \rightarrow \infty} \left(\frac{2\sqrt{n}}{\sqrt{n} \times \sqrt{n}} \right)$$

$$\Rightarrow \lim_{n \rightarrow \infty} \frac{2}{\sqrt{n}}$$

$$\Rightarrow \frac{2}{\sqrt{\infty}}$$

$$\Rightarrow 0$$

107

69) $\lim_{n \rightarrow \infty} 2^{\frac{1}{n}}$

$$\Rightarrow \lim_{n \rightarrow \infty} e^{\ln 2^{\frac{1}{n}}}$$

$$\Rightarrow e^{\lim_{n \rightarrow \infty} \ln 2^{\frac{1}{n}}}$$

————— ①

Now:

$$\lim_{n \rightarrow \infty} \frac{1}{n} \times \ln 2$$

$$\Rightarrow \lim_{n \rightarrow \infty} \frac{\frac{1}{n}}{1}$$

$$\Rightarrow \lim_{n \rightarrow \infty} \frac{1}{n}$$

$$\Rightarrow \frac{1}{\infty}$$

$$= 0$$

form ① $\rightarrow$

$$\lim_{n \rightarrow \infty} = e^0 = 1$$

Handwritten note: ~~Not applicable~~

Optimization Problems

$$\lim_{n \rightarrow \infty} \frac{1}{n} = 0$$

$$\lim_{n \rightarrow \infty} \frac{1}{n} = 0$$

$$\lim_{n \rightarrow \infty} \frac{1}{n} = 0$$

$$\lim_{n \rightarrow \infty} \frac{1}{n} = 0$$

$$\lim_{n \rightarrow \infty} \frac{1}{n} = 0$$

$$\lim_{n \rightarrow \infty} \frac{1}{n} = 0$$

$$\lim_{n \rightarrow \infty} \frac{1}{n} = 0$$

$$\lim_{n \rightarrow \infty} \frac{1}{n} = 0$$

$$\lim_{n \rightarrow \infty} \frac{1}{n} = 0$$

$$\lim_{n \rightarrow \infty} \frac{1}{n} = 0$$

$$\lim_{n \rightarrow \infty} \frac{1}{n} = 0$$

$$\lim_{n \rightarrow \infty} \frac{1}{n} = 0$$

$$\lim_{n \rightarrow \infty} \frac{1}{n} = 0$$

$$\lim_{n \rightarrow \infty} \frac{1}{n} = 0$$

$$\lim_{n \rightarrow \infty} \frac{1}{n} = 0$$

Optimization problems

P-3378

① A box with a square base and open top must have a volume of $32,000 \text{ cm}^3$. Find the dimension of the box that minimize the amount of material used.



$$\therefore V = l \times l \times h$$

$$\Rightarrow l^2 h = 32000$$

$$\Rightarrow h = \frac{32000}{l^2}$$

$$A = l^2 + 4 \times l \times \frac{32000}{l^2}$$

$$A = 4lh + l^2 \rightarrow \text{for open box}$$

Rules:-

1. find the minimum and maximum point of A
 $f'(x) = 0 \rightarrow \text{for SP}$

2. $f''(x) > 0$ minimum

3. $f''(x) < 0$ maximum.

Consider the length, breadth and height as l, l, h

$$\therefore \text{Volume} \Rightarrow l^2 h = 32,000$$

$$\Rightarrow h = \frac{32000}{l^2}$$

$$\therefore \text{Area} \Rightarrow A = l^2 + 4lh$$

$$= l^2 + 4l \times \frac{32000}{l^2}$$

$$= l^2 + 4 \times \frac{32000}{l}$$

We know,
for SP

$$A' = 0$$

$$\Rightarrow 2l + (128000) \times -1 l^{-1-1} = 0$$

$$\Rightarrow 2l - 128000/l^2 = 0$$

$$\Rightarrow 2l - \frac{128000}{l^2} = 0$$

$$\Rightarrow 2l^3 - 128000 = 0$$

$$\Rightarrow l = \sqrt[3]{\frac{128000}{2}} = 40.$$

Find max and minimum:—

$$A' = 2l - \frac{128000}{l^2}$$

$$A'' = 2 - 128000(-2l^{-2-1})$$

$$A'' = 2 + 256000/l^3$$

$$A'' = \frac{2l^3 + 256000}{l^3}$$

Now:-

$A''(40) > 0$ which is positive

So, are the minimum value.